

On the Consequences of Partitioning on Observation: Resolution of Loschmidt's Paradox Through Self-Reference

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Abstract

Loschmidt's paradox claims that reversing all molecular velocities should decrease entropy, apparently violating the Second Law. We prove this reversal is logically impossible through a fundamental temporal paradox.

Loschmidt's procedure requires: (1) time progression exists, so the system can evolve from t_0 to t_T , and (2) time progression can be undone, so the system can return to t_0 . These requirements are contradictory. If time progression exists, then $t_0 \neq t'_0$ (different moments), so reversal fails to return to the initial state. If time progression does not exist, then there is no evolution to reverse. Either way, the procedure fails.

Memory erasure cannot resolve this. If observers erase all memory of temporal progression, they lose the ability to determine their position in time—they cannot tell if they are at t_0 or t'_0 or any other moment. This eliminates the possibility of verifying that reversal succeeded. Moreover, erasure itself is a temporal event that creates distinguishable states: the system before erasure differs from the system after erasure.

The distinguishability is not observer-dependent. States S_0 (at t_0) and S'_0 (at t'_0) occupy different positions in causal structure: S_0 is not causally downstream of the reversal event, while S'_0 is. This causal difference exists independent of any observer and makes the states objectively distinguishable.

We formalize this through partition theory. Observable states exist as intersections of partitions—divisions of reality into actualization/non-actualization pairs. Time creates unavoidable partitions: each moment is distinguished from every other moment by its temporal and causal position. These temporal partitions cannot be deleted without eliminating time itself, which would make evolution (and therefore reversal) meaningless.

We prove: (1) observers are finite and create partitions through biased observation (Section 2), (2) memory erasure eliminates temporal localization (Section 2), (3) Loschmidt's requirements are contradictory (Section 2), (4) causal structure creates observer-independent distinguishability

(Section 2), (5) partitions cannot be deleted (Section 5), (6) enumeration is self-referential (Section 7), and (7) therefore reversal is impossible (Section 8). This establishes the Second Law as a consequence of temporal structure, not statistical mechanics.

Keywords: Loschmidt's paradox, thermodynamic irreversibility, temporal paradox, causality, partition theory, Second Law

1 Introduction

Consider a cup resting on a table. You observe it is on the table because it is *not* on the floor, *not* floating in the air, *not* teleporting across the room. These negations are not philosophical abstractions—they are observable facts. The cup's failure to teleport is as much a fact as its presence on the table.

But why do you observe the cup *on the table* rather than the floor *under the table*, or the air *around the table*, or the atoms *composing the table*? You are a finite observer: you have finite time, finite attention, finite sensory bandwidth. You cannot observe everything simultaneously. You must choose where to look, what to look at, for how long, and at what resolution. This choice is a *bias*, and this bias is the origin of partitioning.

When you look at the cup (rather than the floor), you create the partition {cup, not-cup}. When you look at its location (rather than its molecular composition), you create the partition {on-table, not-on-table}. When you observe for five seconds (rather than ten), you create the partition {observed-for-5s, not-observed-for-5s}. These partitions are not pre-existing features of reality that you discover—they are created by your act of biased observation.

Crucially, *you can never reobserve the same way*. If you observe the cup now and then observe it again one second later, the second observation is fundamentally different from the first:

- **First observation:** You have no prior knowledge of the cup's state. Your bias is unconditioned.
- **Second observation:** You have memory of the first observation. Your bias is conditioned by prior

knowledge.

Even if the cup is in an identical physical state, the observations are different because *you* are different. The observer who has observed once is not the same as the observer who has not yet observed.

This resolves Loschmidt’s paradox [?, 1] through a fundamental temporal contradiction. In 1876, Loschmidt argued that since Newtonian mechanics is time-reversible, one could in principle reverse all molecular velocities in a gas, causing entropy to decrease and apparently violating the Second Law of Thermodynamics. Boltzmann responded with statistical arguments [2]: while reversal is not impossible, the probability of spontaneously arriving at the precise reversed state is astronomically small.

We prove that Loschmidt’s reversal is not merely improbable but *logically impossible*. The proof rests on a simple observation about time:

Loschmidt’s procedure requires:

1. Time progression exists (so the system can evolve from t_0 to t_T)
2. Time progression can be undone (so the system can return to t_0)

These requirements are contradictory:

- If time progression exists, then the moment t_0 (before evolution) is distinguishable from the moment t'_0 (after reversal). They are different moments in the temporal sequence. Therefore, the system at t'_0 has not returned to t_0 —it is merely in a similar configuration at a different time.
- If time progression does not exist, then there is no evolution from t_0 to t_T to reverse. Without temporal progression, entropy cannot increase, so there is nothing to decrease.

Either way, Loschmidt’s reversal fails. This is not a practical limitation—it is a logical impossibility arising from the structure of time itself.

One might object: “Erase the observer’s memory, then the observer cannot tell t_0 from t'_0 , so they become the same.” This fails for a devastating reason: *if the observer erases all memory of temporal progression, the observer loses the ability to determine their position in time*. The observer cannot tell if they are at t_0 (before evolution), t'_0 (after reversal), or any other moment. Without this ability, the observer cannot verify that reversal succeeded. Moreover, the erasure itself is a temporal event—it happens at a specific moment and creates distinguishable states (before-erasure vs. after-erasure).

The key insight is: *If you can tell t_0 from t'_0 , then they are different moments, and reversal has failed. If you cannot tell t_0 from t'_0 , then you have lost temporal localization, and you cannot verify whether reversal*

succeeded. Either way, Loschmidt’s procedure cannot be completed.

This argument is independent of observers. Even in a universe with no observers, the moment t_0 comes before the moment t'_0 in the causal order. The state at t_0 is not causally downstream of the reversal event; the state at t'_0 is causally downstream of the reversal event. This causal difference is intrinsic to spacetime structure and makes the states objectively distinguishable.

We formalize this through *partition theory*. Observable states exist as intersections of partitions—divisions of reality into what-is and what-is-not:

$$\text{Cup-on-table} = \{\text{cup}\} \cap \{\text{on-table}\} \cap \{\text{stationary}\} \cap \{\text{observed-now}\} \cap \dots \quad (1)$$

Time itself creates unavoidable partitions. Each moment is distinguished from every other moment by its temporal position and causal relationships. The partition:

$$\mathcal{P}_{\text{causal}} = \{\text{causally-downstream-of-reversal, not-causally-downstream-of-reversal}\} \quad (2)$$

distinguishes t_0 from t'_0 independent of any observer. These temporal partitions cannot be deleted without eliminating time itself, which would make evolution (and therefore reversal) meaningless.

This resolution differs fundamentally from previous approaches:

- **Boltzmann’s statistical argument** [2]: Reversal is possible but improbable ($\sim e^{-N}$ where $N \sim 10^{23}$). *Our argument*: Reversal is logically impossible due to temporal structure.
- **Szilard’s information-theoretic argument** [?]: Measurement requires information erasure, which generates entropy. *Our argument*: Temporal progression creates irreversible partitions; information erasure cannot eliminate temporal ordering.
- **Landauer’s principle** [5]: Erasing one bit costs $k_B T \ln 2$ in entropy. *Our argument*: Temporal partitions cannot be erased at all—time cannot be undone.
- **Prigogine’s dissipative structures** [?]: Irreversibility emerges from instability and chaos. *Our argument*: Irreversibility is fundamental, arising from the causal structure of spacetime itself.
- **Penrose’s cosmological arrow** [3]: Time asymmetry requires special initial conditions (low-entropy Big Bang). *Our argument*: Time asymmetry arises from the logical structure of temporal progression, independent of initial conditions.

The argument proceeds in four steps:

Step 1: Temporal paradox (Section 2). Loschmidt requires both that time progression exists (for evolution) and that time progression can be undone

(for reversal). These are contradictory. If time exists, then $t_0 \neq t'_0$, so reversal fails. If time doesn't exist, there's nothing to reverse. Memory erasure eliminates temporal localization, making verification impossible.

Step 2: Partitions define observable states (Section 4). Observable states are intersections of partitions created by finite observers through biased observation. Time creates unavoidable partitions: each moment is distinguished by its temporal and causal position. These partitions accumulate irreversibly.

Step 3: Partitions cannot be deleted (Section 5). Deleting a partition deletes the information it encodes. One cannot delete {molecule-here, molecule-not-here} without losing knowledge of the molecule's location. Temporal partitions cannot be deleted without eliminating time itself.

Step 4: Enumeration is self-referential (Section 7). Attempting to enumerate all partitions creates new partitions. Listing "molecule-1 is here" creates {enumerated-molecule-1, not-enumerated-molecule-1}, which itself requires enumeration. This self-reference prevents complete state specification.

Together, these establish that Loschmidt's reversal is impossible, not due to practical limitations but due to the logical structure of time, observation, and causality.

This has profound implications:

1. **Second Law as logical necessity:** Entropy increase is not a statistical tendency that could be violated with sufficient luck or technology. It is a logical necessity arising from temporal structure.
2. **Time asymmetry from causality:** The arrow of time does not require special initial conditions [3] or cosmological boundary conditions [?]. It emerges from the causal ordering of moments, which is intrinsic to spacetime.
3. **Universality of thermodynamics:** All observers—regardless of their physical form, sensory mechanisms, or cognitive architecture—must create partitions to observe. All observers experience the same temporal progression. Therefore, all observers measure the same thermodynamic facts.
4. **Limits of reversible computing:** Even perfectly reversible logic gates [?] cannot reverse observation, because observation creates temporal partitions that cannot be deleted. Reversible computing can minimize entropy generation but cannot eliminate it.
5. **Observer-independence:** While observation creates partitions, temporal and causal structure exist independent of observers. The causal ordering of moments is objective, making thermodynamic irreversibility observer-independent.

The structure of this paper is as follows. Section 2 proves the temporal paradox: Loschmidt's requirements are contradictory, and memory erasure eliminates temporal localization. Section 4 formalizes partitions and proves that observable states are partition intersections. Section 5 proves that partitions cannot be deleted without deleting observed facts. Section 7 proves that partition enumeration is self-referential and cannot complete. Section 8 applies these results to prove that Loschmidt's reversal is impossible. Section 9 derives the entropy formula $S = k_B M \ln n$ from partition structure. Section ?? explains why thermodynamics is universal across all observers. Section ?? connects partition structure to cosmological observations. Section 11 concludes.

Throughout, we emphasize concrete examples over abstract formalism. The cup on the table is not a pedagogical simplification—it is the fundamental case. All thermodynamic systems, from ideal gases to black holes, are observable only through partitions. Understanding the cup is understanding thermodynamics.

2 Finite Observers and the Temporal Paradox

2.1 The Necessity of Bias

An observer is any system that distinguishes states. This definition is deliberately broad: it includes humans, measuring instruments, thermostats, and even simple mechanical systems that respond differently to different inputs. The key property is *finite capacity*: no observer can distinguish all possible states simultaneously.

Consider observing a cup on a table. The cup's complete physical state includes:

- Position: $\mathbf{r} \in \mathbb{R}^3$ (three continuous parameters)
- Orientation: $(\theta, \phi, \psi) \in SO(3)$ (three continuous parameters)
- Molecular configuration: $\{\mathbf{r}_i, \mathbf{v}_i\}_{i=1}^{N_A}$ where $N_A \sim 10^{23}$ (Avogadro's number of positions and velocities)
- Quantum state: $|\psi\rangle \in \mathcal{H}$ where $\dim(\mathcal{H}) \sim 2^{10^{23}}$ (exponentially large Hilbert space)
- Electromagnetic field configuration: $\mathbf{E}(\mathbf{x}), \mathbf{B}(\mathbf{x})$ for all $\mathbf{x}$ (infinite-dimensional field)
- Gravitational field perturbations: $h_{\mu\nu}(\mathbf{x})$ (infinite-dimensional field)
- Thermal fluctuations at all scales
- Quantum vacuum fluctuations

The complete state has infinite information content. No finite observer can access this complete state in finite time. Therefore, observation requires *selection*:

choosing which aspects of the state to observe and which to ignore.

This selection is a *bias*. When you observe “the cup is on the table,” you are biasing your observation toward:

- Spatial location (not molecular composition)
- Macroscopic position (not quantum state)
- Current time (not historical trajectory)
- Visual appearance (not gravitational field)
- Classical properties (not quantum fluctuations)

Different biases yield different observations. An observer biased toward molecular composition would observe “ N_A molecules arranged in cup-shape.” An observer biased toward quantum states would observe “superposition of molecular configurations.” An observer biased toward thermal properties would observe “temperature distribution.” These are not different interpretations of the same observation—they are different observations arising from different biases.

Key point: Bias is not optional. It is not a limitation of current technology or a consequence of human psychology. It is a logical necessity arising from the finiteness of observers. Any system with finite capacity must select which distinctions to make and which to ignore.

2.2 Bias Creates Partitions

Each bias creates a partition: a division of all possible states into two categories based on a single distinguishing feature.

Example 1: Spatial bias. You look at the table (not the floor). This creates the partition:

$$\mathcal{P}_{\text{location}} = \{\text{on-table, not-on-table}\} \quad (3)$$

All possible cup positions are divided into two categories: those on the table and those not on the table. By observing the cup is on the table, you actualize the first category and non-actualize the second.

Example 2: Temporal bias. You observe for 5 seconds (not 10 seconds, not 1 hour). This creates the partition:

$$\mathcal{P}_{\text{duration}} = \{\text{observed-for-5s, not-observed-for-5s}\} \quad (4)$$

All possible observation durations are divided into two categories: 5 seconds and not-5-seconds. By observing for exactly 5 seconds, you actualize the first category.

Example 3: Resolution bias. You observe macroscopic position (not molecular positions). This creates the partition:

$$\mathcal{P}_{\text{resolution}} = \{\text{macroscopic, not-macroscopic}\} \quad (5)$$

All possible levels of detail are divided into two categories: macroscopic (coarse-grained) and microscopic

(fine-grained). By observing macroscopically, you actualize the first category.

Example 4: Categorical bias. You observe the cup as a single object (not as 10^{23} molecules). This creates the partition:

$$\mathcal{P}_{\text{category}} = \{\text{single-object, not-single-object}\} \quad (6)$$

You could have observed the same physical system as a collection of molecules, or as a quantum superposition, or as a thermal distribution. Each choice creates a different partition.

The observable state is the intersection of all actualized categories:

$$\text{Observed state} = \bigcap_i \text{Actualized}(\mathcal{P}_i) \quad (7)$$

For the cup on the table:

$$\text{Cup-on-table} = \{\text{on-table}\} \cap \{\text{observed-for-5s}\} \cap \{\text{macroscopic}\} \cap \{\text{single-object}\} \quad (8)$$

Remove any partition—say, delete the distinction between {on-table, not-on-table}—and the cup’s location becomes undefined. Remove all partitions, and the cup ceases to exist as an observable entity.

2.3 Observation Changes the Observer

Each observation adds partitions to the observer’s state. Before observing the cup, the observer has no partition $\mathcal{P}_{\text{cup}}$. After observing, the observer has:

$$\mathcal{P}_{\text{cup}} = \{\text{observed-cup, not-observed-cup}\} \quad (9)$$

This partition is now part of the observer’s state. The observer who has observed the cup is different from the observer who has not. We denote:

$$\text{Observer}_{\text{before}} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_n\} \quad (10)$$

$$\text{Observer}_{\text{after}} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_n, \mathcal{P}_{\text{cup}}\} \quad (11)$$

Since $\mathcal{P}_{\text{cup}} \notin \text{Observer}_{\text{before}}$, we have:

$$\text{Observer}_{\text{before}} \neq \text{Observer}_{\text{after}} \quad (12)$$

This inequality is irreversible. Once $\mathcal{P}_{\text{cup}}$ is created, it encodes the information that the cup was observed. Deleting this partition would delete this information—the observer would no longer know whether the cup had been observed or not.

2.4 You Cannot Reobserve the Same Way

Suppose you observe the cup at time t_1 , creating partition $\mathcal{P}_1$. Then you observe the cup again at time $t_2 > t_1$. Even if the cup is in an identical physical state, the second observation is different because:

First observation at t_1 :

- Observer state: $\{\mathcal{P}_{\text{prior}}\}$ (no knowledge of cup)
- Bias: Unconditioned (no prior expectations about cup)

- Partition created: $\mathcal{P}_1 = \{\text{cup-at-}t_1, \text{not-cup-at-}t_1\}$
- Observer state after: $\{\mathcal{P}_{\text{prior}}, \mathcal{P}_1\}$

Second observation at t_2 :

- Observer state: $\{\mathcal{P}_{\text{prior}}, \mathcal{P}_1\}$ (has knowledge of cup from t_1)
- Bias: Conditioned by $\mathcal{P}_1$ (expects cup to still be there)
- Partition created: $\mathcal{P}_2 = \{\text{cup-at-}t_2, \text{not-cup-at-}t_2\}$
- Observer state after: $\{\mathcal{P}_{\text{prior}}, \mathcal{P}_1, \mathcal{P}_2\}$

Note that $\mathcal{P}_1 \neq \mathcal{P}_2$ even if the cup is in the same physical state, because:

- $\mathcal{P}_1$ distinguishes cup-at- t_1 from not-cup-at- t_1
- $\mathcal{P}_2$ distinguishes cup-at- t_2 from not-cup-at- t_2
- These are different partitions because they reference different times

Moreover, the observer's bias at t_2 is conditioned by the memory of $\mathcal{P}_1$. The observer expects the cup to still be there, which affects where and how the observer looks. This is a different bias than at t_1 , when the observer had no such expectation.

Concrete example: When you first see the cup, you might scan the entire table to locate it. When you look again, you look directly at the location where you saw it before. These are different observation processes arising from different biases, even though the cup hasn't moved.

2.5 The Indistinguishability Argument

We now prove the central result: if two states are observationally indistinguishable, they are the same state.

Theorem 2.1 (Indistinguishability Implies Identity). *If two states S_1 and S_2 are observationally identical (no observer can distinguish them by any measurement), then they are the same state: $S_1 = S_2$.*

Proof. Observable states are defined by the partitions that distinguish them. A state S is characterized by which categories it belongs to for each partition $\mathcal{P}_i$.

If no observer can distinguish S_1 from S_2 , then for all partitions $\mathcal{P}_i$:

$$S_1 \in \text{Actualized}(\mathcal{P}_i) \iff S_2 \in \text{Actualized}(\mathcal{P}_i) \quad (13)$$

That is, S_1 and S_2 belong to the same category (actualized or non-actualized) for every partition.

Since observable states are intersections of actualized partition categories (Equation 7):

$$S_1 = \bigcap_i \text{Actualized}(\mathcal{P}_i) = S_2 \quad (14)$$

Therefore, S_1 and S_2 are the same intersection, hence the same state. $\square$

The contrapositive is immediate:

Corollary 2.2 (Distinguishability Implies Difference). *If an observer can distinguish states S_1 and S_2 (there exists a partition $\mathcal{P}$ such that S_1 and S_2 belong to different categories), then they are different states: $S_1 \neq S_2$.*

This is not a philosophical claim—it is the definition of “different states” in operational terms. States are different if and only if they can be distinguished by observation.

2.6 Loschmidt's Temporal Paradox

We now apply the indistinguishability argument to Loschmidt's reversal.

Theorem 2.3 (Loschmidt States Are Distinguishable). *The state S_0 at time $t = 0$ (before evolution) is distinguishable from the state S'_0 at time $t = 0'$ (after reversal).*

Proof. Consider the partition:

$$\mathcal{P}_{\text{history}} = \{\text{has-evolved-and-reversed}, \text{has-not-evolved-and-reversed}\} \quad (15)$$

At $t = 0$: The system has not yet evolved to t_T and reversed back. Therefore:

$$S_0 \in \{\text{has-not-evolved-and-reversed}\} \quad (16)$$

At $t = 0'$: The system has evolved to t_T and reversed back. Therefore:

$$S'_0 \in \{\text{has-evolved-and-reversed}\} \quad (17)$$

Since S_0 and S'_0 belong to different categories of partition $\mathcal{P}_{\text{history}}$, they are distinguishable by Corollary 2.2. $\square$

Corollary 2.4 (Loschmidt Reversal Fails). *Since S_0 and S'_0 are distinguishable, they are different states: $S_0 \neq S'_0$. Therefore, Loschmidt's reversal does not return the system to its initial state.*

One might object: “The partition $\mathcal{P}_{\text{history}}$ depends on the observer's memory. If we erase the observer's memory, the partition disappears, and the states become indistinguishable.”

This objection leads to the temporal paradox.

2.7 Memory Erasure Eliminates Temporal Localization

Theorem 2.5 (Memory Erasure Eliminates Temporal Localization). *If an observer erases all memory of temporal progression, the observer cannot determine their position in time.*

Proof. Suppose the observer erases all partitions related to temporal progression:

$$\mathcal{P}_{t_1} = \{\text{observed-at-}t_1, \text{not-observed-at-}t_1\} \quad (\text{erased}) \quad (18)$$

$$\mathcal{P}_{t_2} = \{\text{observed-at-}t_2, \text{not-observed-at-}t_2\} \quad (\text{erased}) \quad (19)$$

$$\vdots \quad (20)$$

$$\mathcal{P}_{t_T} = \{\text{observed-at-}t_T, \text{not-observed-at-}t_T\} \quad (\text{erased}) \quad (21)$$

After erasure, the observer has no partitions that distinguish different moments. Therefore, the observer cannot answer:

- “Am I at t_0 or t'_0 ?” (no partition distinguishes them)
- “Have I observed the cup before?” (no partition records prior observations)
- “Has the system evolved to t_T and reversed?” (no partition records this history)
- “What time is it now?” (no partition identifies the current moment)

Without the ability to answer these questions, the observer cannot locate themselves in temporal progression. From the observer’s perspective, all moments are indistinguishable—the observer exists in a timeless state. $\square$

This creates a devastating problem for verifying Loschmidt’s reversal:

Corollary 2.6 (Reversal Cannot Be Verified After Memory Erasure). *If the observer erases memory to make S_0 and S'_0 indistinguishable, the observer cannot verify that the reversal succeeded.*

Proof. To verify that reversal succeeded, the observer must:

1. Remember the initial state at t_0
2. Observe the current state at t'_0
3. Compare them and confirm they are identical

But if the observer erases memory of the initial state (step 1), they cannot perform the comparison (step 2) or confirmation (step 3). The observer cannot know whether they have returned to the initial state or are at some other state entirely. $\square$

Moreover, erasure itself creates new distinguishability:

Theorem 2.7 (Erasure Creates New Partitions). *Memory erasure is itself an observation that creates new partitions, preventing return to the initial state.*

Proof. Memory erasure occurs at a specific moment in time—say, t_{erase} . This creates the partition:

$$\mathcal{P}_{\text{erase}} = \{\text{memory-erased-at-}t_{\text{erase}}, \text{memory-not-erased-at-}t_{\text{erase}}\} \quad (22)$$

At t_0 (before evolution): Memory has not been erased. Therefore:

$$S_0 \in \{\text{memory-not-erased-at-}t_{\text{erase}}\} \quad (23)$$

At t'_0 (after reversal and erasure): Memory has been erased. Therefore:

$$S'_0 \in \{\text{memory-erased-at-}t_{\text{erase}}\} \quad (24)$$

Since S_0 and S'_0 belong to different categories of partition $\mathcal{P}_{\text{erase}}$, they remain distinguishable. The erasure operation itself prevents return to the initial state. $\square$

Analogy: Consider two sheets of paper:

- Sheet A: Blank, never written on
- Sheet B: Had writing, which was then erased

These are distinguishable. Sheet B has:

- Indentations from pen pressure
- Eraser marks and smudges
- Slightly rougher surface texture
- Chemical residues from ink and eraser

Similarly, a system that has evolved and reversed (with memory erased) is distinguishable from a system that never evolved. The erasure leaves traces.

2.8 The Fundamental Temporal Paradox

We now prove the central result: Loschmidt’s procedure requires contradictory assumptions about time.

Theorem 2.8 (Loschmidt’s Temporal Paradox). *Loschmidt’s reversal requires both:*

1. *Time progression exists (so the system can evolve from t_0 to t_T)*
2. *Time progression can be undone (so the system can return to t_0)*

These requirements are contradictory.

Proof. Requirement 1 (Time progression exists):

For the system to evolve from t_0 to t_T , the moments $t_0, t_1, \dots, t_T$ must be distinguishable. If they were not distinguishable, they would be the same moment (by Theorem 2.1), and no evolution would occur. Therefore:

$$t_i \neq t_j \quad \text{for all } i \neq j \quad (25)$$

In particular, $t_0 \neq t'_0$ where t'_0 is the moment after reversal when the system supposedly “returns” to

the initial state. These are different moments in the temporal sequence.

Requirement 2 (Time progression can be undone):

For the system to “return” to the initial state, the moment t'_0 must be identical to t_0 . Otherwise, the system has not returned—it is merely in a similar configuration at a different moment. Therefore:

$$t'_0 = t_0 \quad (26)$$

Contradiction:

From Requirement 1: $t_0 \neq t'_0$

From Requirement 2: $t'_0 = t_0$

These cannot both be true. Therefore, Loschmidt’s reversal is logically contradictory. $\square$

Corollary 2.9 (Dilemma for Loschmidt). *Loschmidt faces an inescapable dilemma:*

- **If time progression exists:** Then $t_0 \neq t'_0$, so reversal does not return to the initial state (Corollary 2.4).
- **If time progression does not exist:** Then there is no evolution from t_0 to t_T , so there is nothing to reverse.

Either way, the reversal procedure fails.

2.9 Observer-Independent Distinguishability

One might object that the preceding arguments rely on observer memory—a “subjective” element. We now prove that the distinguishability of S_0 and S'_0 is observer-independent, arising from the causal structure of spacetime itself.

Definition 2.10 (Causal Partition). For any event E , define the causal partition:

$$\mathcal{P}_E = \{\text{causally-downstream-of-}E, \text{not-causally-downstream-of-}E\} \quad (27)$$

A state S at spacetime point x is causally downstream of E at point y if y is in the past light cone of x .

Theorem 2.11 (Causal Distinguishability). *Let E_{reversal} be the event of velocity reversal at time t_T . Then:*

$$S_0 \in \{\text{not-causally-downstream-of-}E_{\text{reversal}}\} \quad (28)$$

$$S'_0 \in \{\text{causally-downstream-of-}E_{\text{reversal}}\} \quad (29)$$

Therefore, $S_0 \neq S'_0$ independent of any observer.

Proof. The state S_0 at time t_0 occurs before the reversal event E_{reversal} at time t_T (since $t_0 < t_T$ by assumption). Therefore, E_{reversal} is not in the past light cone of S_0 . Hence:

$$S_0 \in \{\text{not-causally-downstream-of-}E_{\text{reversal}}\} \quad (30)$$

The state S'_0 at time t'_0 occurs after the reversal event E_{reversal} at time t_T (since the system must evolve

from t_T back to t'_0 , so $t'_0 > t_T > t_0$ in the timeline). Therefore, E_{reversal} is in the past light cone of S'_0 . Hence:

$$S'_0 \in \{\text{causally-downstream-of-}E_{\text{reversal}}\} \quad (31)$$

Since S_0 and S'_0 belong to different categories of the causal partition $\mathcal{P}_{E_{\text{reversal}}}$, they are different states by Corollary 2.2. This holds independent of any observer—it is a fact about the causal structure of spacetime. $\square$

Key point: Causal structure exists whether or not anyone observes it. The fact that E_{reversal} is in the past light cone of S'_0 but not of S_0 is an objective feature of spacetime geometry. This makes the distinguishability of S_0 and S'_0 observer-independent.

2.10 Time-Stamping Is Unavoidable

The fundamental reason $S_0 \neq S'_0$ is that *time itself creates partitions*. Every moment in time is distinguishable from every other moment. This is not a limitation of clocks or measurement—it is the definition of time as a dimension.

Consider: How do you know it is now t_2 rather than t_1 ? Because you can distinguish them. If you could not distinguish t_1 from t_2 —if no measurement, no observation, no physical process could tell them apart—then they would be the same moment by Theorem 2.1. The distinguishability of moments is what makes time a dimension.

Each moment creates a partition:

$$\mathcal{P}_{t_i} = \{\text{at-time-}t_i, \text{not-at-time-}t_i\} \quad (32)$$

These partitions are not created by observers—they are intrinsic to the structure of time. Any event at time t_i belongs to the category $\{\text{at-time-}t_i\}$, and this category is distinct from $\{\text{at-time-}t_j\}$ for $i \neq j$.

Therefore:

- State at t_0 : Belongs to category $\{\text{at-time-}t_0\}$
- State at t'_0 : Belongs to category $\{\text{at-time-}t'_0\}$

Since $t_0 \neq t'_0$ (they are different moments in the timeline by Requirement 1 of Theorem 2.8), the categories are different, and therefore the states are different:

$$S_0 \in \{\text{at-time-}t_0\} \neq \{\text{at-time-}t'_0\} \ni S'_0 \quad (33)$$

One might object: “But t_0 and t'_0 have the same coordinate value in the time dimension. If we reset the clock to show $t = 0$ at both moments, they are the same time.”

This confuses *coordinate labels* with *temporal identity*. Consider a spatial analogy:

- You walk from point A (coordinates $x = 0$) to point B (coordinates $x = 10$)

- Then you walk back to point A (coordinates $x = 0$ again)

You are at the same spatial coordinates, but these are not the same event. The event “arriving at A after visiting B” is distinguishable from the event “starting at A before visiting B.” They occur at different times and have different causal relationships to the journey.

Similarly:

- At t_0 : System has coordinate $t = 0$, has not evolved to t_T
- At t'_0 : System has coordinate $t = 0$ (after resetting clock), has evolved to t_T and reversed

These are distinguishable events. The coordinate label is the same, but the temporal and causal positions are different.

2.11 The Only Way Loschmidt Could Work

For Loschmidt’s reversal to truly return the system to its initial state, time progression would have to not exist at all. The moments t_0 and t'_0 would have to be literally the same moment—not just similar, but identical in the sense that there is no temporal distance between them, no causal ordering, no distinction whatsoever.

This would require:

- No causal ordering (no before/after relationships)
- No temporal progression (no flow of time)
- All moments simultaneous (block universe with no temporal dimension)
- No evolution (no change from one moment to another)

But if time progression does not exist, then:

- The system cannot evolve from t_0 to t_T (no temporal progression to evolve through)
- There is no “reversal” to perform (no temporal direction to reverse)
- Entropy cannot increase or decrease (entropy is defined by temporal progression of macrostates)
- Thermodynamics is meaningless (the Second Law is a statement about temporal evolution)

Therefore, the only universe in which Loschmidt’s reversal could work is a universe in which there is nothing to reverse. The reversal procedure is self-defeating: it requires time progression to have something to reverse, but time progression is precisely what prevents the reversal from succeeding.

Theorem 2.12 (Loschmidt’s Ultimate Impossibility). *Loschmidt’s reversal is impossible in any universe with temporal progression, and meaningless in any universe without temporal progression.*

Proof. Case 1: Universe has temporal progression.

Then moments are distinguishable by their temporal position and causal relationships (Theorem 2.11). In particular, $t_0 \neq t'_0$. Therefore, reversal does not return to the initial state (Corollary 2.4).

Case 2: Universe has no temporal progression.

Then there is no evolution from t_0 to t_T , because evolution requires temporal progression—a sequence of distinguishable moments through which the system passes. Without evolution, there is no entropy increase to reverse. The reversal procedure is meaningless because there is nothing to reverse.

In both cases, Loschmidt’s reversal fails. $\square$

2.12 Summary: The Temporal Paradox

We have established:

1. **Distinguishability defines difference:** States are different if and only if they can be distinguished by observation (Theorem 2.1).
2. **Loschmidt states are distinguishable:** S_0 and S'_0 can be distinguished by their history, causal relationships, and temporal positions (Theorem 2.3, Theorem 2.11).
3. **Memory erasure eliminates temporal localization:** If observers erase memory to make states indistinguishable, they lose the ability to determine their position in time (Theorem 2.5).
4. **Erase creates new partitions:** The act of erasing memory creates new distinguishability, preventing return to the initial state (Theorem 2.7).
5. **Temporal paradox:** Loschmidt requires both that time progression exists (for evolution) and that time progression can be undone (for reversal). These are contradictory (Theorem 2.8).
6. **Observer-independent:** The distinguishability arises from causal structure, which exists independent of observers (Theorem 2.11).
7. **Ultimate impossibility:** Reversal is impossible in universes with time, and meaningless in universes without time (Theorem 2.12).

The key insight is: *As long as time progression is included, reversal cannot work.* Time progression creates an irreversible ordering of moments through causal structure. This ordering is not observer-dependent—it is intrinsic to spacetime geometry. Loschmidt’s reversal attempts to undo this ordering, but the ordering is what makes the reversal procedure meaningful in the first place. The procedure is self-defeating.

This completes the proof that Loschmidt’s reversal is logically impossible. The remaining sections will develop the partition formalism in more detail and explore its consequences for entropy, thermodynamics, and cosmology.

3 The Two Components of Entropy

We begin by distinguishing two fundamentally different contributions to total entropy. This distinction is essential for understanding why Loschmidt's reversal fails: velocity inversion addresses only one component, leaving the other irreversible.

3.1 Kinetic Entropy

Classical thermodynamic entropy, as formulated by Boltzmann [2], measures the multiplicity of microstates consistent with a given macrostate:

$$S_{\text{kinetic}} = k_B \ln \Omega$$

where Ω is the number of accessible microstates and $k_B = 1.380649 \times 10^{-23}$ J/K is Boltzmann's constant.

This entropy quantifies energy dispersal across degrees of freedom. For an ideal gas of N particles in volume V at temperature T , the number of accessible microstates grows with phase space volume:

$$\Omega \sim \left(\frac{V}{h^3}\right)^N (2\pi m k_B T)^{3N/2}$$

where h is Planck's constant and m is particle mass. Kinetic entropy increases through processes that spread energy across available degrees of freedom: diffusion (spatial spreading), heat conduction (temperature equilibration), and viscous dissipation (conversion of ordered motion to random motion).

3.1.1 Formal Reversibility of Kinetic Entropy

Kinetic entropy is formally reversible under time inversion. The Hamiltonian equations of motion:

$$\dot{\mathbf{q}}_i = \frac{\partial H}{\partial \mathbf{p}_i}, \quad \dot{\mathbf{p}}_i = -\frac{\partial H}{\partial \mathbf{q}_i}$$

are invariant under the transformation $t \rightarrow -t$, $\mathbf{p}_i \rightarrow -\mathbf{p}_i$ (with $\mathbf{q}_i$ unchanged). If $(\mathbf{q}(t), \mathbf{p}(t))$ is a solution, then so is $(\mathbf{q}(-t), -\mathbf{p}(-t))$.

Therefore, if a system evolves from low-entropy state $\mathbf{x}_0$ to high-entropy state $\mathbf{x}_f$ along trajectory γ , the time-reversed trajectory γ^* (with all velocities inverted) evolves from $\mathbf{x}'_f = (\mathbf{q}_f, -\mathbf{p}_f)$ back to $\mathbf{x}_0$, decreasing kinetic entropy from $S_{\text{kinetic}}(\mathbf{x}_f)$ to $S_{\text{kinetic}}(\mathbf{x}_0)$.

This is the basis of Loschmidt's paradox: the microscopic dynamics permit entropy decrease, yet macroscopically we never observe it.

3.2 Categorical Entropy

We now introduce a second entropy component arising from categorical distinctions—the partitioning of state space into distinguishable categories.

Definition 3.1 (Categorical Entropy). For a system with M degrees of freedom, each capable of n distinguishable states, the categorical entropy is:

$$S_{\text{categorical}} = k_B M \ln n$$

This entropy measures the information content of state distinctions rather than energy dispersal. Each degree of freedom contributes $k_B \ln n$ to the total entropy by virtue of being distinguishable from $n - 1$ alternative states.

3.2.1 Physical Origin of Categorical Entropy

Categorical entropy arises from partition operations—the division of state space into distinguishable regions. Consider a gas in a container divided by an imaginary plane into left and right halves. Even if the gas has uniform density and temperature (maximum kinetic entropy), we can still distinguish particles on the left from particles on the right. This distinction contributes to categorical entropy.

More generally, any observable that takes discrete values partitions state space. For a spin- $\frac{1}{2}$ particle, the spin observable partitions state space into $\{|\uparrow\rangle, |\downarrow\rangle\}$, contributing $k_B \ln 2$ to categorical entropy. For N such particles, the total categorical entropy is:

$$S_{\text{categorical}} = N k_B \ln 2$$

This entropy is independent of the particles' spatial distribution or kinetic energies. It exists purely by virtue of the spin states being distinguishable.

3.2.2 Categorical Entropy in Classical Systems

In classical mechanics, categorical entropy arises from the discretization of continuous observables. Although position and momentum are formally continuous, physical measurements have finite resolution δq and δp . This divides phase space into cells of volume $(\delta q \cdot \delta p)^{3N}$.

For a particle confined to region of size L with momentum resolution δp , the number of distinguishable states is:

$$n \sim \frac{L}{\delta q} \cdot \frac{p_{\text{max}}}{\delta p}$$

The categorical entropy is:

$$S_{\text{categorical}} = k_B \ln n = k_B \ln \left(\frac{L \cdot p_{\text{max}}}{\delta q \cdot \delta p} \right)$$

This entropy increases when:

- The accessible region L expands (spatial partition refinement)
- The momentum range p_{max} increases (momentum partition refinement)
- The measurement resolution improves (δq or δp decreases)

Crucially, categorical entropy can increase even when kinetic entropy is constant. If a gas at equilibrium (maximum kinetic entropy) is observed with progressively finer spatial resolution, categorical entropy increases because more distinctions become accessible.

3.3 Independence of the Two Entropies

Kinetic and categorical entropy are independent in the sense that each can change while the other remains constant.

Example 1: Kinetic entropy increases, categorical entropy constant.

A gas initially confined to one half of a container expands to fill the entire volume. Kinetic entropy increases as particles spread spatially. However, if we maintain the same partition structure (e.g., tracking only whether particles are in the left or right half), categorical entropy remains constant at $Nk_B \ln 2$.

Example 2: Categorical entropy increases, kinetic entropy constant.

A gas at thermal equilibrium (maximum kinetic entropy) is observed with a finer spatial grid. The number of distinguishable positions increases from n_1 to $n_2 > n_1$, so categorical entropy increases from $k_B M \ln n_1$ to $k_B M \ln n_2$. Kinetic entropy remains at its maximum value $k_B \ln \Omega_{\max}$.

Example 3: Both increase simultaneously.

A gas expands and is simultaneously observed with finer resolution. Both kinetic and categorical entropy increase.

3.4 Total Entropy

The total entropy of a physical system is the sum of kinetic and categorical contributions:

$$S_{\text{total}} = S_{\text{kinetic}} + S_{\text{categorical}}$$

More explicitly:

$$S_{\text{total}} = k_B \ln \Omega + k_B M \ln n$$

where Ω is the number of microstates (energy dispersal) and $M \ln n$ quantifies the number of categorical distinctions (partition structure).

3.5 Why Loschmidt Reversal Cannot Decrease Total Entropy

Loschmidt's velocity reversal $\mathbf{p}_i \rightarrow -\mathbf{p}_i$ addresses only kinetic entropy. Under this transformation:

$$S_{\text{kinetic}}(\mathbf{x}_f) \rightarrow S_{\text{kinetic}}(\mathbf{x}_0) \quad (\text{decreases})$$

However, categorical entropy is unaffected by velocity reversal:

$$S_{\text{categorical}}(\mathbf{x}_f) \rightarrow S_{\text{categorical}}(\mathbf{x}_f) \quad (\text{unchanged})$$

This is because categorical entropy depends on partition structure, not on particle velocities. Reversing velocities does not erase the distinctions that have been made.

Therefore, total entropy after reversal is:

$$S_{\text{total}}^{\text{reversed}} = S_{\text{kinetic}}(\mathbf{x}_0) + S_{\text{categorical}}(\mathbf{x}_f)$$

Compare this to the initial total entropy:

$$S_{\text{total}}^{\text{initial}} = S_{\text{kinetic}}(\mathbf{x}_0) + S_{\text{categorical}}(\mathbf{x}_0)$$

The difference is:

$$\Delta S_{\text{total}} = S_{\text{total}}^{\text{reversed}} - S_{\text{total}}^{\text{initial}} = S_{\text{categorical}}(\mathbf{x}_f) - S_{\text{categorical}}(\mathbf{x}_0)$$

Since categorical entropy increases monotonically (as we will prove in Section ??), we have:

$$S_{\text{categorical}}(\mathbf{x}_f) > S_{\text{categorical}}(\mathbf{x}_0)$$

Therefore:

$$\Delta S_{\text{total}} > 0$$

Conclusion: Even if kinetic entropy could be perfectly reversed, total entropy does not decrease because categorical entropy remains irreversible. This provides a partial resolution of Loschmidt's paradox, but leaves open the question of *why* categorical entropy is irreversible. We address this in the next section.

3.6 Relation to Information-Theoretic Entropy

Categorical entropy is closely related to Shannon entropy [9]. For a system with n equally probable states, Shannon entropy is:

$$H = - \sum_{i=1}^n p_i \ln p_i = -n \cdot \frac{1}{n} \ln \frac{1}{n} = \ln n$$

Multiplying by k_B and summing over M degrees of freedom yields:

$$S_{\text{categorical}} = k_B M H = k_B M \ln n$$

However, there is a crucial difference: Shannon entropy measures uncertainty about which state the system is in, while categorical entropy measures the number of distinctions that have been made, regardless of uncertainty.

Even if we know with certainty that the system is in state i (so $p_i = 1$ and Shannon entropy is zero), categorical entropy is non-zero because the state i is distinguished from $n - 1$ alternatives. The distinction itself carries information, independent of our knowledge.

This distinction becomes important when analyzing Loschmidt's reversal: even with perfect knowledge of the microstate (zero Shannon entropy), categorical entropy remains non-zero and irreversible.

3.7 Summary

We have established:

1. Total entropy comprises two independent components: kinetic entropy (energy dispersal) and categorical entropy (partition structure).
2. Kinetic entropy is formally reversible under velocity inversion, but categorical entropy is not.

3. Loschmidt's reversal can at most decrease kinetic entropy, leaving categorical entropy unchanged, so total entropy does not decrease.
4. Categorical entropy measures the number of distinctions made, not the uncertainty about the system's state.

The next section addresses the mechanism by which categorical entropy is generated and why it is irreversible.

4 Partitions as Observable Structure

4.1 What Is a Partition?

A partition is a division of all possible states into two mutually exclusive and exhaustive categories. Every partition has the form:

$$\mathcal{P} = \{A, \neg A\} \quad (34)$$

where A is some property and $\neg A$ is its negation (not- A). The two categories satisfy:

$$A \cap \neg A = \emptyset \quad (\text{mutually exclusive}) \quad (35)$$

$$A \cup \neg A = \Omega \quad (\text{exhaustive, where } \Omega \text{ is the space of all possible states}) \quad (36)$$

Every state belongs to exactly one category: either A or $\neg A$, never both, never neither.

Examples:

- $\mathcal{P}_{\text{location}} = \{\text{on-table, not-on-table}\}$
- $\mathcal{P}_{\text{motion}} = \{\text{moving, not-moving}\}$
- $\mathcal{P}_{\text{temperature}} = \{\text{hot, not-hot}\}$
- $\mathcal{P}_{\text{existence}} = \{\text{exists, not-exists}\}$

Note that partitions are binary: they divide reality into two categories, not three or more. This is not a limitation—any multi-way distinction can be decomposed into multiple binary partitions.

Example: The distinction “cup is red, blue, or green” decomposes into:

$$\mathcal{P}_{\text{red}} = \{\text{red, not-red}\} \quad (37)$$

$$\mathcal{P}_{\text{blue}} = \{\text{blue, not-blue}\} \quad (38)$$

$$\mathcal{P}_{\text{green}} = \{\text{green, not-green}\} \quad (39)$$

A red cup actualizes $\{\text{red}\}$ and non-actualizes $\{\text{not-blue}\}$ and $\{\text{not-green}\}$.

4.2 Actualization and Non-Actualization

When a partition $\mathcal{P} = \{A, \neg A\}$ is applied to a state S , exactly one category is *actualized* (the state belongs to it) and the other is *non-actualized* (the state does not belong to it).

Definition: For a state S and partition $\mathcal{P} = \{A, \neg A\}$:

$$\text{Actualized}(\mathcal{P}, S) = \begin{cases} A & \text{if } S \in A \\ \neg A & \text{if } S \in \neg A \end{cases} \quad (40)$$

Example: Cup on table.

- Partition: $\mathcal{P}_{\text{location}} = \{\text{on-table, not-on-table}\}$
- State: Cup is on table
- Actualized: $\{\text{on-table}\}$
- Non-actualized: $\{\text{not-on-table}\}$

The actualized category is what the state *is*. The non-actualized category is what the state *is not*. Both are observable facts. The cup's failure to be on the floor is as much a fact as its presence on the table.

4.3 States as Partition Intersections

An observable state is not a primitive entity—it is defined by the partitions that distinguish it. Specifically, a state is the intersection of all its actualized partition categories.

Definition 4.1 (Observable State). An observable state S is the intersection of the actualized categories of all partitions that apply to it:

$$S = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i, S) \quad (41)$$

where $\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_M\}$ is the set of all partitions that distinguish S .

Example: Cup on table.

$$\text{Cup-on-table} = \{\text{cup}\} \cap \{\text{on-table}\} \cap \{\text{stationary}\} \cap \{\text{ceramic}\} \cap \{\text{white}\} \quad (42)$$

$$\cap \{\text{cylindrical}\} \cap \{\text{room-temperature}\} \cap \{\text{solid}\} \cap \{\text{visible}\} \quad (43)$$

Each partition contributes one constraint. The state is the set of all configurations that satisfy all constraints simultaneously.

Key insight: Remove any partition, and the state becomes less defined. Remove all partitions, and the state ceases to exist as an observable entity.

Theorem 4.2 (Partition Dependence). *An observable state cannot exist without partitions. If all partitions are removed, the state becomes undefined.*

Proof. Suppose we remove all partitions: $M = 0$. Then by Definition 4.1:

$$S = \bigcap_{i=1}^0 \text{Actualized}(\mathcal{P}_i, S) = \Omega \quad (44)$$

where Ω is the space of all possible states (the empty intersection is the universal set).

But $S = \Omega$ means the state is “everything”—it does not distinguish any particular configuration from any other. This is not an observable state; it is the absence of any distinction. Therefore, observable states require partitions. $\square$

4.4 Partitions Are Not Subjective

One might think partitions are subjective—that different observers create different partitions based on their interests or perspectives. This is incorrect. Partitions encode objective facts about what is and what is not.

Example: Consider the partition:

$$\mathcal{P}_{\text{teleport}} = \{\text{cup-teleports}, \text{cup-does-not-teleport}\} \quad (45)$$

Every observer—human, hypothetical sentient cow, alien intelligence, or automated measuring device—will observe the same actualization:

$$\text{Actualized}(\mathcal{P}_{\text{teleport}}) = \{\text{cup-does-not-teleport}\} \quad (46)$$

This is not a matter of perspective or interpretation. The cup’s failure to teleport is an objective fact enforced by the laws of physics. All observers observe the same non-actualization.

What differs between observers: Which partitions they choose to apply. A human might apply:

$$\mathcal{P}_{\text{human}} = \{\text{on-table}, \text{not-on-table}\} \quad (47)$$

while a molecular physicist might apply:

$$\mathcal{P}_{\text{physicist}} = \{\text{molecules-in-cup-configuration}, \text{molecules-not-in-cup-configuration}\} \quad (48)$$

These are different partitions, but both encode objective facts. The human observes the macroscopic location; the physicist observes the molecular configuration. These are different observations of the same underlying reality, not different interpretations of the same observation.

Theorem 4.3 (Partition Objectivity). *For any partition $\mathcal{P}$ and state S , all observers agree on which category is actualized: $\text{Actualized}(\mathcal{P}, S)$ is observer-independent.*

Proof. Suppose two observers O_1 and O_2 apply the same partition $\mathcal{P} = \{A, \neg A\}$ to the same state S .

If O_1 observes $S \in A$ and O_2 observes $S \in \neg A$, then they have made different observations. But by assumption, they applied the same partition to the same state. Therefore, they must have performed different measurements or observed at different times—they did not actually apply the same partition to the same state.

If they truly apply the same partition to the same state (same measurement, same time, same conditions), they must observe the same actualization. This is what it means for the state to be objective: all observers agree on its properties when they perform the same measurement. $\square$

Key point: Objectivity does not mean observer-independence in the sense that states exist without observation. It means inter-observer agreement: all observers who apply the same partition to the same state observe the same result.

4.5 Continuous vs. Discrete Partitions

So far, we have considered discrete partitions like $\{\text{on-table}, \text{not-on-table}\}$. But many physical properties are continuous (position, momentum, energy). How do partitions apply to continuous variables?

Answer: Continuous variables are observed through discretization—dividing the continuous range into bins.

Example: Position $x \in \mathbb{R}$.

To observe position, we must ask: “Is the cup in region R or not in region R ?” This creates the partition:

$$\mathcal{P}_R = \{\text{in-}R, \text{not-in-}R\} \quad (49)$$

Different choices of region R create different partitions:

$$\mathcal{P}_{\text{table}} = \{\text{on-table}, \text{not-on-table}\} \quad (R = \text{table surface}) \quad (50)$$

$$\mathcal{P}_{\text{left}} = \{\text{left-half}, \text{right-half}\} \quad (R = \{x < 0\}) \quad (51)$$

$$\mathcal{P}_{\text{bin-}i} = \{\text{in-bin-}i, \text{not-in-bin-}i\} \quad (R = [x_i, x_i + \Delta x]) \quad (52)$$

To specify position to precision Δx , we need $N = L/\Delta x$ partitions, where L is the size of the observable region. Each partition distinguishes one bin from all others.

Key insight: Continuous variables require infinite partitions for exact specification. Finite observers can only apply finite partitions, so they observe continuous variables with finite precision.

Theorem 4.4 (Finite Precision). *A finite observer applying M partitions can distinguish at most 2^M states.*

Proof. Each partition $\mathcal{P}_i$ has two categories: $\{A_i, \neg A_i\}$. The observer determines which category is actualized for each partition. This gives M binary choices, hence 2^M possible combinations of actualized categories.

Each combination corresponds to a distinct observable state:

$$S = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \quad (53)$$

Therefore, the observer can distinguish at most 2^M states. $\square$

Example: To distinguish N positions along a line requires $M = \log_2 N$ partitions.

- $M = 1$ partition: distinguishes 2 positions (left vs. right)
- $M = 2$ partitions: distinguishes 4 positions (left-left, left-right, right-left, right-right)
- $M = 10$ partitions: distinguishes $2^{10} = 1024$ positions
- $M = 23$ partitions: distinguishes $2^{23} \approx 10^7$ positions (roughly 1 cm precision in a room)

4.6 Partition Hierarchies

Partitions can be organized hierarchically: coarse partitions divide the state space into large regions, while fine partitions subdivide those regions further.

Example: Position of cup.

Level 1 (coarse):

$$\mathcal{P}_{\text{room}} = \{\text{in-room}, \text{not-in-room}\} \quad (54)$$

Level 2 (medium):

$$\mathcal{P}_{\text{table}} = \{\text{on-table}, \text{not-on-table}\} \quad (55)$$

(This only makes sense if cup is in-room; if not-in-room, the table partition is irrelevant)

Level 3 (fine):

$$\mathcal{P}_{\text{left-half}} = \{\text{left-half-of-table}, \text{right-half-of-table}\} \quad (56)$$

(This only makes sense if cup is on-table)

Level 4 (very fine):

$$\mathcal{P}_{\text{bin-}i} = \{\text{in-bin-}i, \text{not-in-bin-}i\} \quad (57)$$

(This specifies position to precision Δx)

The hierarchy reflects the structure of observation: we first determine coarse properties (is it in the room?), then refine (is it on the table?), then refine further (which part of the table?), and so on.

Key insight: Each level of the hierarchy adds partitions. Coarser observations use fewer partitions; finer observations use more partitions. The total number of partitions determines the precision of the observable state.

4.7 Partition Algebra

Partitions can be combined using logical operations.

Intersection (AND): Given partitions $\mathcal{P}_1 = \{A_1, \neg A_1\}$ and $\mathcal{P}_2 = \{A_2, \neg A_2\}$, their intersection creates four categories:

$$\mathcal{P}_1 \cap \mathcal{P}_2 = \{A_1 \cap A_2, A_1 \cap \neg A_2, \neg A_1 \cap A_2, \neg A_1 \cap \neg A_2\} \quad (58)$$

This is not a single partition (which must have exactly two categories), but a *partition structure*—a set of partitions that jointly distinguish four states.

Example:

$$\mathcal{P}_{\text{table}} = \{\text{on-table}, \text{not-on-table}\} \quad (59)$$

$$\mathcal{P}_{\text{moving}} = \{\text{moving}, \text{not-moving}\} \quad (60)$$

Their intersection distinguishes four states:

- On-table and moving
- On-table and not-moving
- Not-on-table and moving
- Not-on-table and not-moving

Union (OR): Given partitions $\mathcal{P}_1 = \{A_1, \neg A_1\}$ and $\mathcal{P}_2 = \{A_2, \neg A_2\}$, their union creates the partition:

$$\mathcal{P}_1 \cup \mathcal{P}_2 = \{A_1 \cup A_2, \neg(A_1 \cup A_2)\} = \{A_1 \cup A_2, \neg A_1 \cap \neg A_2\} \quad (61)$$

This distinguishes states where at least one property holds from states where neither holds.

Example:

$$\mathcal{P}_{\text{table}} = \{\text{on-table}, \text{not-on-table}\} \quad (62)$$

$$\mathcal{P}_{\text{floor}} = \{\text{on-floor}, \text{not-on-floor}\} \quad (63)$$

Their union:

$$\mathcal{P}_{\text{table}} \cup \mathcal{P}_{\text{floor}} = \{\text{on-table-or-floor}, \text{neither-table-nor-floor}\} \quad (64)$$

Negation (NOT): Given partition $\mathcal{P} = \{A, \neg A\}$, its negation swaps the categories:

$$\neg \mathcal{P} = \{\neg A, A\} \quad (65)$$

This is the same partition with categories relabeled. Negation does not create new distinctions—it only changes which category is considered “positive.”

4.8 Partition Refinement

A partition $\mathcal{P}_2$ is a *refinement* of partition $\mathcal{P}_1$ if $\mathcal{P}_2$ makes all the distinctions that $\mathcal{P}_1$ makes, plus additional distinctions.

Example:

$$\mathcal{P}_1 = \{\text{in-room}, \text{not-in-room}\} \quad (66)$$

$$\mathcal{P}_2 = \{\text{on-table}, \text{not-on-table}\} \quad (67)$$

If the table is in the room, then $\mathcal{P}_2$ refines $\mathcal{P}_1$: knowing the cup is on-table implies it is in-room. The partition $\mathcal{P}_2$ makes a finer distinction within the in-room category.

Formally:

Definition 4.5 (Partition Refinement). Partition $\mathcal{P}_2 = \{A_2, \neg A_2\}$ refines partition $\mathcal{P}_1 = \{A_1, \neg A_1\}$ if:

$$A_2 \subseteq A_1 \quad \text{or} \quad A_2 \subseteq \neg A_1 \quad (68)$$

That is, one category of $\mathcal{P}_2$ is entirely contained within one category of $\mathcal{P}_1$.

Refinement creates a partial order on partitions: $\mathcal{P}_1 \preceq \mathcal{P}_2$ means $\mathcal{P}_2$ refines $\mathcal{P}_1$ (makes finer distinctions).

Key insight: Observation proceeds by refinement. We start with coarse partitions (is it in the room?) and progressively refine (is it on the table? which part of the table? exact position?). Each refinement adds information by making finer distinctions.

4.9 Information Content of Partitions

Each partition encodes one bit of information: the answer to a yes/no question.

Theorem 4.6 (Partition Information). *A partition $\mathcal{P} = \{A, \neg A\}$ encodes exactly 1 bit of information: which category is actualized.*

Proof. The partition has two categories. Observing which category is actualized requires distinguishing between two possibilities. This is the definition of 1 bit of information. $\square$

Therefore, a state defined by M partitions encodes M bits of information.

Corollary 4.7 (State Information). *An observable state defined by M partitions encodes M bits of information.*

Example: To specify the position of a cup on a table to 1 cm precision:

- Table dimensions: 1 m $\times$ 1 m
- Number of 1 cm bins: $100 \times 100 = 10^4$
- Number of partitions needed: $M = \log_2(10^4) \approx 13.3 \approx 14$ bits

These 14 partitions encode 14 bits of information about the cup's position.

4.10 Partitions and Entropy

The connection to thermodynamic entropy is now clear. Entropy measures the number of partitions required to specify a macrostate.

Traditional definition: Boltzmann entropy:

$$S = k_B \ln \Omega \quad (69)$$

where Ω is the number of microstates compatible with the macrostate.

Partition definition: If M partitions are required to specify the macrostate, then:

$$\Omega = 2^M \quad (70)$$

(each partition doubles the number of distinguishable states). Therefore:

$$S = k_B \ln(2^M) = k_B M \ln 2 \quad (71)$$

Defining M as the number of partitions, we have:

$$S = k_B M \ln 2 \quad (72)$$

This is the entropy formula in terms of partitions. We will derive this more carefully in Section 9.

Key insight: Entropy is not a property of the microstate—it is a property of the partition structure used to observe the macrostate. Different partition structures (different ways of coarse-graining) yield different entropies for the same microstate.

4.11 Example: Gas in a Box

Consider an ideal gas: N molecules in a box of volume V .

Microstate: Specified by positions and momenta of all molecules:

$$\{\mathbf{r}_1, \mathbf{p}_1, \mathbf{r}_2, \mathbf{p}_2, \dots, \mathbf{r}_N, \mathbf{p}_N\} \quad (73)$$

This requires $6N$ continuous parameters (3 position + 3 momentum per molecule).

Macrostate: Specified by coarse-grained properties:

- Total energy E (one partition: $\{E \in [E_0, E_0 + \Delta E], E \notin [E_0, E_0 + \Delta E]\}$)
- Total volume V (one partition: $\{V = V_0, V \neq V_0\}$)
- Number of particles N (one partition: $\{N = N_0, N \neq N_0\}$)

But this is insufficient to specify the macrostate precisely. We also need:

- Spatial distribution: How many molecules in each region of the box?
- Momentum distribution: How many molecules with each momentum?

Spatial partitions: Divide the box into n cells. For each cell i , create partition:

$$\mathcal{P}_i = \{\text{molecule-in-cell-}i, \text{molecule-not-in-cell-}i\} \quad (74)$$

This requires $\log_2 n$ partitions per molecule, or $N \log_2 n$ partitions total.

Momentum partitions: Divide momentum space into m cells. For each cell j , create partition:

$$\mathcal{P}_j = \{\text{momentum-in-cell-}j, \text{momentum-not-in-cell-}j\} \quad (75)$$

This requires $\log_2 m$ partitions per molecule, or $N \log_2 m$ partitions total.

Total partitions:

$$M = N \log_2 n + N \log_2 m = N \log_2(nm) \quad (76)$$

Entropy:

$$S = k_B M \ln 2 = k_B N \log_2(nm) \ln 2 = k_B N \ln(nm) \quad (77)$$

If we choose $n = V/v_0$ (where v_0 is a quantum cell volume) and $m = p^3/p_0^3$ (where p_0 is a momentum scale), we recover the Sackur-Tetrode equation for ideal gas entropy.

Key point: The entropy depends on how finely we partition position and momentum space. Finer partitions (smaller cells) require more partitions, yielding higher entropy. This is why entropy is not an intrinsic property of the microstate—it depends on the partition structure used to observe it.

4.12 Partitions vs. Measurements

One might think partitions are the same as measurements. They are closely related but not identical.

Measurement: A physical process that determines which category of a partition is actualized.

Partition: The logical structure of the distinction being made (the division into categories).

Example: Consider the partition:

$$\mathcal{P}_{\text{hot}} = \{\text{temperature} > 50^\circ\text{C}, \text{temperature} \leq 50^\circ\text{C}\} \quad (78)$$

This partition exists as a logical structure independent of any particular measurement. Different measurements can determine which category is actualized:

- Thermometer reading
- Thermal expansion of a material
- Infrared camera image
- Human touch sensation

These are different measurements, but they all determine the same partition. The partition is the distinction; the measurement is the physical process that determines which side of the distinction the state falls on.

Key insight: Partitions are observer-independent (all observers agree on which category is actualized), but measurement-dependent (different measurements can determine the same partition). The partition structure is objective; the choice of which partitions to apply is subjective (depends on the observer's interests and capabilities).

4.13 Summary: Partitions as Observable Structure

We have established:

1. **Partitions are binary divisions:** Every partition divides states into two categories: $\{A, \neg A\}$.
2. **States are partition intersections:** Observable states are defined by the intersection of actualized partition categories (Definition 4.1).
3. **Partition dependence:** States cannot exist without partitions (Theorem 4.2).
4. **Partition objectivity:** All observers agree on which category is actualized for a given partition and state (Theorem 4.3).
5. **Finite precision:** A finite observer applying M partitions can distinguish at most 2^M states (Theorem 4.4).
6. **Partition hierarchies:** Partitions can be organized hierarchically from coarse to fine, with each level adding more partitions.

7. **Partition information:** Each partition encodes 1 bit of information (Theorem 4.6); a state defined by M partitions encodes M bits (Corollary 4.7).

8. **Entropy as partition count:** Entropy is proportional to the number of partitions required to specify the macrostate (Equation 72).

This partition formalism provides the foundation for understanding thermodynamic irreversibility. In the next section, we prove that partitions cannot be deleted without deleting the information they encode, establishing the fundamental irreversibility of observation.

5 Irreversibility of Partition Deletion

5.1 What Does It Mean to Delete a Partition?

A partition $\mathcal{P} = \{A, \neg A\}$ encodes the distinction between states that have property A and states that do not. To delete this partition means to eliminate this distinction—to make it impossible to tell whether a state has property A or not.

Example: Consider the partition:

$$\mathcal{P}_{\text{location}} = \{\text{on-table}, \text{not-on-table}\} \quad (79)$$

To delete this partition means:

- You can no longer tell whether the cup is on the table or not
- The distinction between on-table and not-on-table ceases to exist
- All information about the cup's location relative to the table is lost

This is not the same as forgetting where the cup is. Forgetting means you once knew but no longer remember. Deleting the partition means the distinction itself is eliminated—it becomes impossible for anyone to determine the cup's location, even in principle.

5.2 Partition Deletion Deletes Information

Theorem 5.1 (Partition Deletion Deletes Information). *Deleting a partition $\mathcal{P}$ deletes the information encoded by $\mathcal{P}$. If $\mathcal{P} = \{A, \neg A\}$ is deleted, it becomes impossible to determine whether a state has property A or not.*

Proof. By Theorem 4.6, partition $\mathcal{P}$ encodes 1 bit of information: which category is actualized.

Before deletion:

- If state $S \in A$, we know S has property A
- If state $S \in \neg A$, we know S does not have property A

After deletion of $\mathcal{P}$:

- The distinction between A and $\neg A$ no longer exists

- We cannot determine whether $S \in A$ or $S \in \neg A$
- The information about which category S belongs to is lost

Therefore, deleting $\mathcal{P}$ deletes 1 bit of information. $\square$

Concrete example: Suppose the cup is on the table. This means:

$$\text{Actualized}(\mathcal{P}_{\text{location}}) = \{\text{on-table}\} \quad (80)$$

This is a fact—an observed property of the current state. If we delete partition $\mathcal{P}_{\text{location}}$, we delete this fact. We can no longer say the cup is on the table, because the distinction between on-table and not-on-table no longer exists.

5.3 You Cannot Delete a Partition Without Deleting the State

Recall from Definition 4.1 that an observable state is the intersection of actualized partition categories:

$$S = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \quad (81)$$

Each partition contributes one constraint to the definition of the state. Remove a partition, and the state becomes less defined.

Theorem 5.2 (Partition Deletion Changes State). *Deleting a partition $\mathcal{P}_j$ from a state S defined by partitions $\{\mathcal{P}_1, \dots, \mathcal{P}_M\}$ produces a different state S' defined by partitions $\{\mathcal{P}_1, \dots, \mathcal{P}_{j-1}, \mathcal{P}_{j+1}, \dots, \mathcal{P}_M\}$.*

Proof. Before deletion:

$$S = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) = \text{Actualized}(\mathcal{P}_j) \cap \left(\bigcap_{i \neq j} \text{Actualized}(\mathcal{P}_i) \right) \quad (82)$$

After deletion of $\mathcal{P}_j$:

$$S' = \bigcap_{i \neq j} \text{Actualized}(\mathcal{P}_i) \quad (83)$$

Since S includes the constraint from $\mathcal{P}_j$ and S' does not:

$$S \subseteq S' \quad (84)$$

That is, S' is a larger set than S —it includes all states in S plus additional states that satisfy the remaining partitions but not $\mathcal{P}_j$. Therefore, $S \neq S'$. $\square$

Example: Cup on table.

$$S = \{\text{cup}\} \cap \{\text{on-table}\} \cap \{\text{stationary}\} \quad (85)$$

Delete the partition $\mathcal{P}_{\text{location}}$ = {on-table, not-on-table}:

$$S' = \{\text{cup}\} \cap \{\text{stationary}\} \quad (86)$$

Now S' includes:

- Cup on table and stationary (the original state S)
- Cup on floor and stationary (not in S)
- Cup floating in air and stationary (not in S)
- Cup anywhere and stationary (not in S)

The state S' is less specific than S . We have lost the information that the cup is on the table.

5.4 Partition Deletion Is Irreversible

Theorem 5.3 (Irreversibility of Partition Deletion). *Once a partition $\mathcal{P}$ is deleted, it cannot be recovered without new observation. The information encoded by $\mathcal{P}$ is permanently lost.*

Proof. Suppose partition $\mathcal{P} = \{A, \neg A\}$ is deleted from state S . Before deletion, we knew whether $S \in A$ or $S \in \neg A$. After deletion, this information is lost.

To recover $\mathcal{P}$, we must determine which category S belongs to. But this requires observation: we must measure whether S has property A or not. This measurement creates a new partition $\mathcal{P}'$ (which may be the same distinction as $\mathcal{P}$, but is a new observation).

Crucially, we cannot recover the original partition $\mathcal{P}$ without new observation. The information that was encoded in $\mathcal{P}$ (which category was actualized) is gone. Even if we create a new partition $\mathcal{P}'$ with the same logical structure, we do not know whether the new observation yields the same result as the deleted observation.

Example: Suppose the cup was on the table, and we delete the partition $\mathcal{P}_{\text{location}}$. Now we no longer know whether the cup is on the table. To recover this information, we must observe the cup again. But in the time between deletion and re-observation, the cup might have moved. The new observation might yield {on-floor} instead of {on-table}. We cannot recover the original information without re-observing, and re-observation might yield a different result.

Therefore, partition deletion is irreversible: the original information is permanently lost. $\square$

5.5 Loschmidt's Reversal Requires Partition Deletion

We now apply these results to Loschmidt's paradox. Loschmidt's procedure requires reversing the system from state S_T (at time t_T) back to state S_0 (at time t_0). This requires deleting all partitions created during the evolution from t_0 to t_T .

Theorem 5.4 (Loschmidt Requires Partition Deletion). *To reverse a system from state S_T to state S_0 , all partitions created during evolution from t_0 to t_T must be deleted.*

Proof. At time t_0 , the system is in state:

$$S_0 = \bigcap_{i=1}^{M_0} \text{Actualized}(\mathcal{P}_i) \quad (87)$$

defined by M_0 partitions.

During evolution from t_0 to t_T , new partitions are created:

- Partitions recording observations at intermediate times: $\mathcal{P}_{t_1}, \mathcal{P}_{t_2}, \dots, \mathcal{P}_{t_{T-1}}$
- Partitions recording the system's trajectory
- Partitions recording interactions with the environment
- Partitions recording the observer's memory of the evolution

At time t_T , the system is in state:

$$S_T = \bigcap_{i=1}^{M_T} \text{Actualized}(\mathcal{P}_i) \quad (88)$$

where $M_T > M_0$ (more partitions exist at t_T than at t_0).

To return to state S_0 , we must have:

$$S_0 = \bigcap_{i=1}^{M_0} \text{Actualized}(\mathcal{P}_i) \quad (89)$$

with exactly the same M_0 partitions as at the initial time.

This requires deleting all partitions created during evolution:

$$\{\mathcal{P}_{M_0+1}, \mathcal{P}_{M_0+2}, \dots, \mathcal{P}_{M_T}\} \quad (90)$$

Therefore, Loschmidt's reversal requires partition deletion. $\square$

Corollary 5.5 (Loschmidt Requires Information Deletion). *Loschmidt's reversal requires deleting $(M_T - M_0)$ bits of information—the information encoded by the partitions created during evolution.*

Proof. By Theorem 4.6, each partition encodes 1 bit of information. Deleting $(M_T - M_0)$ partitions deletes $(M_T - M_0)$ bits of information. $\square$

5.6 You Cannot Delete What You Have Observed

The fundamental problem is that partitions encode observed facts. Once you have observed that the cup is on the table, this fact exists. To delete the partition $\mathcal{P}_{\text{location}}$ is to delete this fact—to make it as if the observation never occurred.

Theorem 5.6 (Observed Facts Cannot Be Deleted). *Once a partition $\mathcal{P}$ has been actualized (an observation has been made), the fact of this actualization cannot be deleted without deleting the observation itself.*

Proof. Suppose partition $\mathcal{P} = \{A, \neg A\}$ has been actualized, with result $S \in A$. This means:

- An observation was performed

- The observation determined that S has property A
- This determination is a fact: $S \in A$

To delete partition $\mathcal{P}$ means to eliminate the distinction between A and $\neg A$. After deletion:

- The fact that $S \in A$ no longer exists
- It is no longer true that S has property A (because the property A is no longer defined)
- The observation that determined $S \in A$ is retroactively undone

But this is impossible. The observation occurred. It was a physical process that happened at a specific time and place. The observation cannot be retroactively undone—it can only be forgotten (observer erases memory) or contradicted (new observation yields different result).

Forgetting does not delete the partition—it only deletes the observer's memory of which category was actualized. The partition itself (the distinction between A and $\neg A$) still exists.

Contradiction does not delete the partition—it creates a new partition at a later time that may yield a different result. The original partition and its actualization still exist as historical facts.

Therefore, observed facts cannot be deleted. Once an observation has occurred, the partition it creates exists permanently. $\square$

Analogy: Suppose you write “The cup is on the table” in a notebook. This statement is now a fact—it exists as ink on paper. You can:

- Erase the ink (analogous to forgetting)
- Write a new statement “The cup is on the floor” (analogous to new observation)
- Burn the notebook (analogous to destroying the observer)

But you cannot make it so that the statement was never written. The act of writing occurred. It was a physical process that happened. Even if you erase the ink, the fact that you once wrote the statement remains (there are indentations in the paper, chemical residues, your memory of writing it, etc.).

Similarly, once an observation has occurred, it cannot be undone. The observation was a physical process. Even if you delete the observer's memory, the fact that the observation occurred remains encoded in the causal structure of spacetime.

5.7 Partition Deletion Violates Causality

Theorem 5.7 (Partition Deletion Violates Causality). *Deleting a partition created by an observation at time t_1 requires retroactively changing the past, which violates causality.*

Proof. Suppose an observation at time t_1 creates partition $\mathcal{P}$ with actualization $S \in A$. This observation is a causal event: it has causes (the state of the system at t_1 , the observer's decision to observe, etc.) and effects (the observer's memory, records of the observation, correlations with the environment, etc.).

At time $t_2 > t_1$, suppose we attempt to delete partition $\mathcal{P}$. This requires:

1. Eliminating the distinction between A and $\neg A$
2. Eliminating the fact that the observation at t_1 determined $S \in A$
3. Eliminating all effects of the observation (memory, records, correlations)

But the observation at t_1 is in the past of t_2 . It is causally upstream. To eliminate it requires changing the past—making it so that the observation at t_1 never occurred.

In special relativity, events in the past light cone cannot be changed. The past is fixed by causality. Therefore, deleting a partition created by a past observation violates causality. $\square$

Concrete example: At time $t_1 = 0$, you observe the cup is on the table. This creates partition $\mathcal{P}_{\text{location}}$ with actualization {on-table}. This observation is now in the past.

At time $t_2 = 10$ seconds, you attempt to delete $\mathcal{P}_{\text{location}}$. This requires making it so that the observation at $t_1 = 0$ never occurred. But the observation at $t_1 = 0$ is in your past light cone. It is a fixed event in spacetime. You cannot change it.

You can forget the observation (erase your memory), but this does not delete the observation itself. The observation occurred. It was a physical event. It has left traces in the causal structure of spacetime (photons bounced off the cup and entered your eyes, neurons fired in your brain, etc.). These traces cannot be eliminated without violating causality.

5.8 Partition Deletion Requires Infinite Energy

Even if we ignore causality, partition deletion requires reversing all physical processes that contributed to creating the partition. This requires infinite energy in the thermodynamic limit.

Theorem 5.8 (Partition Deletion Requires Infinite Energy). *Deleting a partition that has been recorded in N physical systems requires energy at least $E \geq Nk_B T \ln 2$ (Landauer's bound). For macroscopic observations, $N \sim 10^{23}$, making deletion practically impossible.*

Proof. When an observation creates partition $\mathcal{P}$, the result is recorded in multiple physical systems:

- Observer's memory (neurons, synapses)
- Photons scattered from the observed object

- Thermal fluctuations in the environment
- Quantum entanglement with surrounding systems

Each recording is a physical bit. By Landauer's principle [5], erasing one bit requires dissipating energy:

$$E_{\text{bit}} = k_B T \ln 2 \quad (91)$$

For a macroscopic observation, the partition is recorded in $N \sim 10^{23}$ systems (roughly Avogadro's number—the observation involves macroscopic amounts of matter). Therefore, erasing the partition requires:

$$E_{\text{total}} = Nk_B T \ln 2 \sim 10^{23} k_B T \ln 2 \quad (92)$$

At room temperature ($T \sim 300$ K):

$$E_{\text{total}} \sim 10^{23} \times 1.38 \times 10^{-23} \times 300 \times 0.693 \sim 3 \times 10^2 \text{ J} \sim 300 \text{ J} \quad (93)$$

This is the energy to erase one partition from one macroscopic observation. For Loschmidt's reversal, we must erase all partitions created during evolution from t_0 to t_T . If observations occur continuously, this is an infinite number of partitions, requiring infinite energy.

Even if observations occur discretely (say, once per second for T seconds), we must erase T partitions, requiring:

$$E_{\text{Loschmidt}} \sim T \times 300 \text{ J} \quad (94)$$

For $T = 1$ hour = 3600 seconds:

$$E_{\text{Loschmidt}} \sim 10^6 \text{ J} \sim 1 \text{ MJ} \quad (95)$$

This is the energy required just to erase the observer's memory of the evolution. To erase all environmental records, correlations, and entanglements requires vastly more energy—likely comparable to the total energy of the system being reversed. $\square$

Key point: Partition deletion is not just logically problematic—it is energetically prohibitive. The energy required to erase macroscopic observations exceeds the energy available in most realistic scenarios.

5.9 Partition Deletion and the Second Law

We can now state the connection to the Second Law of Thermodynamics.

Theorem 5.9 (Partition Accumulation Implies Entropy Increase). *If partitions accumulate irreversibly (cannot be deleted), then entropy increases irreversibly.*

Proof. By Equation 72, entropy is proportional to the number of partitions:

$$S = k_B M \ln 2 \quad (96)$$

If partitions accumulate (new partitions are created but old partitions cannot be deleted), then M increases over time:

$$M(t_2) > M(t_1) \quad \text{for } t_2 > t_1 \quad (97)$$

Therefore:

$$S(t_2) = k_B M(t_2) \ln 2 > k_B M(t_1) \ln 2 = S(t_1) \quad (98)$$

Entropy increases over time. This is the Second Law of Thermodynamics. $\square$

Corollary 5.10 (Second Law as Logical Necessity). *The Second Law of Thermodynamics is a logical consequence of the irreversibility of partition deletion, not a statistical tendency.*

Proof. By Theorem 5.3, partition deletion is impossible (information cannot be recovered without new observation). By Theorem 5.9, irreversible partition accumulation implies entropy increase. Therefore, entropy increase is a logical necessity, not a statistical tendency. $\square$

Key insight: The Second Law is often presented as a statistical principle: entropy is likely to increase because high-entropy states are more numerous than low-entropy states. But this misses the fundamental point. Entropy *must* increase because partitions *cannot* be deleted. This is not a matter of probability—it is a matter of logic.

5.10 Why Boltzmann’s Statistical Argument Is Incomplete

Boltzmann argued that Loschmidt’s reversal is possible in principle but improbable in practice. The probability of spontaneously arriving at the precise reversed state is:

$$P \sim e^{-\Delta S/k_B} \sim e^{-N} \quad (99)$$

where $N \sim 10^{23}$ is the number of particles. This probability is astronomically small but nonzero.

Our argument shows this is incorrect. Loschmidt’s reversal is not merely improbable—it is impossible. The reason is not statistical (low probability) but logical (partition deletion violates information conservation, causality, and energy conservation).

Boltzmann’s error: He assumed that the microstate contains all information about the system. If you know the exact positions and momenta of all molecules, you can in principle reverse them. But this ignores the partitions created by observation. The microstate alone is not sufficient—you must also account for the partition structure that defines the observable macrostate.

Correct view: The observable state is not just the microstate—it is the microstate plus the partition structure. To reverse the observable state, you must reverse both:

1. Reverse the microstate (reverse all molecular velocities)
2. Delete all partitions created during evolution

Step 1 is possible in principle (though practically impossible due to sensitivity to initial conditions). Step 2 is impossible in principle (violates information conservation, causality, and energy conservation). Therefore, Loschmidt’s reversal is impossible.

5.11 Partition Deletion vs. Memory Erasure

One might object: “We can erase memory. Doesn’t this delete partitions?”

Answer: Memory erasure deletes the observer’s record of which category was actualized, but it does not delete the partition itself.

Example: Suppose you observe the cup is on the table, creating partition:

$$\mathcal{P}_{\text{location}} = \{\text{on-table, not-on-table}\} \quad (100)$$

with actualization {on-table}.

If you erase your memory:

- You no longer remember that the cup is on the table
- But the partition $\mathcal{P}_{\text{location}}$ still exists (the distinction between on-table and not-on-table still exists)
- The cup is still on the table (the actualization still holds)
- You just don’t remember it

Memory erasure is like closing your eyes. The world doesn’t disappear—you just can’t see it anymore.

To truly delete the partition would require:

- Eliminating the distinction between on-table and not-on-table
- Making it impossible for anyone (not just you) to determine the cup’s location
- Erasing all records, traces, and correlations that encode the cup’s location

This is far stronger than memory erasure. It requires erasing the information from the entire universe—from the photons that bounced off the cup, the air molecules that were displaced by it, the gravitational field perturbations it created, the quantum entanglements with surrounding atoms, etc.

By Theorem 5.8, this requires energy $E \sim N k_B T \ln 2$ where N is the number of systems that recorded the information. For macroscopic observations, $N \sim 10^{23}$, making deletion practically impossible.

5.12 Partition Deletion and Time’s Arrow

The irreversibility of partition deletion is intimately connected to the arrow of time.

Theorem 5.11 (Partition Deletion Requires Time Reversal). *To delete all partitions created during evolution from t_0 to t_T requires reversing time itself, not just reversing molecular velocities.*

Proof. Partitions are created by observations at specific times:

- $\mathcal{P}_{t_1}$: Partition created by observation at time t_1

- $\mathcal{P}_{t_2}$: Partition created by observation at time t_2
- $\vdots$
- $\mathcal{P}_{t_T}$: Partition created by observation at time t_T

Each partition encodes not just what was observed, but when it was observed. The partition $\mathcal{P}_{t_1}$ is distinct from $\mathcal{P}_{t_2}$ even if they make the same distinction, because they occur at different times.

To delete these partitions requires making it so that the observations at times $t_1, t_2, \dots, t_T$ never occurred. This requires reversing time itself—going back to before the observations happened.

But reversing time is not the same as reversing molecular velocities. Reversing molecular velocities changes the direction of motion but not the direction of time. Time still flows forward; the molecules just move backward. To truly reverse time requires changing the causal structure of spacetime, which is impossible within the framework of relativity. $\square$

Key insight: Loschmidt’s reversal conflates two different concepts:

1. **Dynamical reversal:** Reversing molecular velocities (changing $\mathbf{v} \rightarrow -\mathbf{v}$)
2. **Temporal reversal:** Reversing time itself (changing $t \rightarrow -t$)

Dynamical reversal is possible in principle (though practically impossible). Temporal reversal is impossible in principle (violates causality). Loschmidt’s procedure requires both, but only considers the first. This is why it fails.

5.13 Summary: Irreversibility of Partition Deletion

We have established:

1. **Partition deletion deletes information:** Deleting a partition deletes 1 bit of information (Theorem 5.1).
2. **Partition deletion changes state:** Deleting a partition produces a different, less-defined state (Theorem 5.2).
3. **Partition deletion is irreversible:** Once deleted, information cannot be recovered without new observation (Theorem 5.3).
4. **Loschmidt requires partition deletion:** Reversing from S_T to S_0 requires deleting all partitions created during evolution (Theorem 5.4).
5. **Observed facts cannot be deleted:** Once an observation has occurred, it cannot be retroactively undone (Theorem 5.6).
6. **Partition deletion violates causality:** Deleting a past observation requires changing the past, which violates causality (Theorem 5.7).

7. **Partition deletion requires infinite energy:** Erasing macroscopic observations requires energy $\sim Nk_B T \ln 2$ where $N \sim 10^{23}$ (Theorem 5.8).
8. **Partition accumulation implies entropy increase:** Irreversible partition accumulation implies irreversible entropy increase (Theorem 5.9).
9. **Second Law as logical necessity:** The Second Law is a logical consequence of partition irreversibility, not a statistical tendency (Corollary 5.10).
10. **Memory erasure is insufficient:** Erasing observer memory does not delete partitions—it only deletes the record of which category was actualized.
11. **Partition deletion requires time reversal:** Deleting temporal partitions requires reversing time itself, not just molecular velocities (Theorem 5.11).

The key insight is: *You cannot delete what you have observed.* Observation creates partitions, and partitions encode facts. Facts cannot be deleted—they can only be forgotten (observer erases memory) or contradicted (new observation yields different result). But forgetting and contradiction do not restore the initial state; they create new states with additional partitions. Therefore, observation is irreversible, and entropy increases.

6 The Structure of Non-Actualisations

We now formalize the concept of non-actualisations—states that could have occurred but did not—and analyze their role in specifying the actual state of a physical system. This analysis is crucial for understanding why Loschmidt’s reversal fails: the reversal requires complete knowledge of the current state, which in turn requires enumerating all non-actualisations.

6.1 Actualisations and Non-Actualisations

Definition 6.1 (Actualisation). An actualisation is a state that occurs in physical reality. If a system evolves from initial state $\mathbf{x}_0$ at time $t = 0$ to final state $\mathbf{x}_f$ at time $t = t_f$ along trajectory γ , then all states $\mathbf{x}(t) \in \gamma$ for $t \in [0, t_f]$ are actualisations.

Definition 6.2 (Dynamically Accessible State). A state $\mathbf{y}$ is dynamically accessible from initial state $\mathbf{x}_0$ if there exists a trajectory γ' satisfying the equations of motion such that $\gamma'(0) = \mathbf{x}_0$ and $\mathbf{y} \in \gamma'$.

Definition 6.3 (Non-Actualisation). A non-actualisation is a state that is dynamically accessible from the initial conditions but did not occur. The set of non-actualisations at time t is:

$$\mathcal{N}(t) = \mathcal{S}_{\text{accessible}}(t) \setminus \{\mathbf{x}(t)\}$$

where $\mathcal{S}_{\text{accessible}}(t)$ is the set of all states dynamically accessible at time t , and $\mathbf{x}(t)$ is the actual state.

Example 1: Classical particle. A particle is released from height h and falls to the ground. The actualisation is the trajectory $\mathbf{x}(t) = (x(t), y(t), v_x(t), v_y(t))$ satisfying Newton’s equations. Non-actualisations include:

- Trajectories with different initial velocities (if initial velocity had uncertainty)
- Trajectories where the particle bounces elastically
- Trajectories where the particle stops at intermediate heights

All these are dynamically possible (satisfy $F = ma$) but did not happen.

Example 2: Quantum measurement. A spin- $\frac{1}{2}$ particle in state $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ is measured along the z -axis. Suppose the outcome is $|\uparrow\rangle$. Then:

- Actualisation: $|\uparrow\rangle$
- Non-actualisation: $|\downarrow\rangle$

The state $|\downarrow\rangle$ was dynamically possible (had probability $|\beta|^2$) but did not occur.

Example 3: Gas expansion. A gas confined to the left half of a container is released. It expands to fill the entire volume. The actualisation is the trajectory where molecules spread uniformly. Non-actualisations include:

- All molecules remaining in the left half
- All molecules moving to the right half
- Molecules arranging in a crystal lattice

These are dynamically possible (satisfy Hamiltonian dynamics) but have vanishingly small probability and did not occur.

6.2 Non-Actualisations as Categorical Facts

A crucial observation: non-actualisations are not merely absent—they are *facts* about what did not happen.

When the particle falls and hits the ground, the statement “the particle did not bounce” becomes a permanent fact about the universe. This fact has information content: it eliminates one possibility from the set of potential outcomes.

Definition 6.4 (Categorical Fact). A categorical fact is a proposition of the form “state X did not occur” that becomes permanently true once the actualisation is determined.

Categorical facts are irreversible. Once it is true that “the particle did not bounce”, no subsequent evolution can make this statement false. The particle may bounce in the future, but that does not change the fact that it did not bounce at time t .

6.3 Information Content of Non-Actualisations

The information content of an actualisation is determined by the number of non-actualisations it excludes.

To say “the system is in state $\mathbf{x}$ ” is equivalent to saying “the system is not in state $\mathbf{y}$ ” for all $\mathbf{y} \neq \mathbf{x}$. The more non-actualisations are excluded, the more information is conveyed by specifying the actualisation.

Theorem 6.5 (Information Content of Actualisation). *The information content of an actualisation $\mathbf{x}$ is:*

$$I(\mathbf{x}) = k_B \ln |\mathcal{N}|$$

where $|\mathcal{N}|$ is the number (or measure) of non-actualisations excluded by $\mathbf{x}$.

Proof. The actualisation $\mathbf{x}$ selects one state from the set of accessible states $\mathcal{S}_{\text{accessible}} = \{\mathbf{x}\} \cup \mathcal{N}$. The number of alternatives excluded is $|\mathcal{N}|$.

By Shannon’s information theory [9], the information gained by learning that the system is in state $\mathbf{x}$ rather than any of the $|\mathcal{N}|$ alternatives is:

$$I(\mathbf{x}) = \ln(|\mathcal{N}| + 1) \approx \ln |\mathcal{N}|$$

for $|\mathcal{N}| \gg 1$.

Multiplying by k_B converts from nats to thermodynamic entropy units. $\square$ $\square$

Implication: To specify the actualisation $\mathbf{x}$ completely, we must know all elements of $\mathcal{N}$. We must enumerate which states did not occur.

6.4 The Enumeration Problem

Loschmidt’s reversal requires complete specification of the system’s current state $\mathbf{x}(t)$. This is necessary to determine which velocities to reverse.

However, specifying $\mathbf{x}(t)$ requires distinguishing it from all other possible states. This is equivalent to enumerating $\mathcal{N}(t)$ —the set of all non-actualisations at time t .

Why enumeration is required:

Consider a system with N particles. The phase space has dimension $6N$. To specify a point $\mathbf{x}(t) = (\mathbf{q}_1, \dots, \mathbf{q}_N, \mathbf{p}_1, \dots, \mathbf{p}_N)$ in this space, we must provide $6N$ real numbers.

But real numbers have infinite precision. To specify $\mathbf{q}_i$ to precision δq requires $\ln(L/\delta q)$ bits of information, where L is the size of the accessible region. For N particles in 3D, the total information required is:

$$I_{\text{total}} = 6N \ln(L/\delta q)$$

This information is precisely the information needed to exclude all non-actualisations: we must specify which of the $(L/\delta q)^{6N}$ possible states the system is *not* in.

Therefore, specifying the actualisation is equivalent to enumerating the non-actualisations.

6.5 Non-Actualisations Accumulate Irreversibly

Each time the system evolves, new non-actualisations are created.

At time t , the system is in state $\mathbf{x}(t)$. All other accessible states form $\mathcal{N}(t)$.

At time $t + dt$, the system is in state $\mathbf{x}(t + dt)$. The set of non-actualisations is now:

$$\mathcal{N}(t+dt) = \mathcal{N}(t) \cup \{\text{new states that became accessible but did not occur}\}$$

The set $\mathcal{N}$ grows monotonically:

$$|\mathcal{N}(t + dt)| \geq |\mathcal{N}(t)|$$

This is because:

1. Old non-actualisations remain non-actualisations (they did not occur in the past, and this fact is permanent).
2. New non-actualisations are created as the system evolves (new alternative trajectories become distinguishable).

Theorem 6.6 (Monotonic Growth of Non-Actualisations). *The number of non-actualisations increases monotonically with time:*

$$\frac{d|\mathcal{N}|}{dt} \geq 0$$

Proof. Consider the system at times t and $t + dt$. At time t , the actual state is $\mathbf{x}(t)$, and the non-actualisations are $\mathcal{N}(t)$.

At time $t + dt$, the system has evolved to $\mathbf{x}(t + dt)$. The previous state $\mathbf{x}(t)$ is now in the past. Any alternative state $\mathbf{y}(t) \in \mathcal{N}(t)$ remains a non-actualisation—it did not occur at time t , and this fact is permanent.

Additionally, at time $t + dt$, there are new accessible states that did not occur. These form the increment:

$$\Delta\mathcal{N} = \{\mathbf{y}(t+dt) : \mathbf{y}(t+dt) \text{ is accessible but } \mathbf{y}(t+dt) \neq \mathbf{x}(t+dt)\}$$

Therefore:

$$\mathcal{N}(t + dt) = \mathcal{N}(t) \cup \Delta\mathcal{N}$$

Since $\Delta\mathcal{N}$ is non-empty (there are always alternative trajectories), we have:

$$|\mathcal{N}(t + dt)| > |\mathcal{N}(t)|$$

Taking the limit $dt \rightarrow 0$:

$$\frac{d|\mathcal{N}|}{dt} \geq 0$$

□

□

6.6 Connection to Categorical Entropy

The monotonic growth of non-actualisations is directly related to the increase of categorical entropy.

From Theorem 4.1, the information content of the actualisation is:

$$I(\mathbf{x}(t)) = k_B \ln |\mathcal{N}(t)|$$

This is precisely the categorical entropy:

$$S_{\text{categorical}}(t) = I(\mathbf{x}(t)) = k_B \ln |\mathcal{N}(t)|$$

By Theorem 4.2:

$$\frac{dS_{\text{categorical}}}{dt} = k_B \frac{d \ln |\mathcal{N}|}{dt} = k_B \frac{1}{|\mathcal{N}|} \frac{d|\mathcal{N}|}{dt} \geq 0$$

Therefore, categorical entropy increases monotonically because non-actualisations accumulate irreversibly.

6.7 Non-Actualisations in Phase Space

Geometrically, non-actualisations correspond to the complement of the actual trajectory in phase space.

The actual trajectory is a one-dimensional curve γ in $6N$ -dimensional phase space:

$$\gamma = \{\mathbf{x}(t) : t \in [0, t_f]\}$$

The non-actualisations at time t are all points in the accessible region except $\mathbf{x}(t)$:

$$\mathcal{N}(t) = \mathcal{S}_{\text{accessible}}(t) \setminus \{\mathbf{x}(t)\}$$

The volume of $\mathcal{N}(t)$ is:

$$|\mathcal{N}(t)| = \int_{\mathcal{S}_{\text{accessible}}(t)} d^{6N}x - 0 = V_{\text{accessible}}(t)$$

where the actual trajectory has measure zero (it is a one-dimensional curve in $6N$ dimensions).

As the system evolves, the accessible region expands (due to diffusion, energy dispersal, etc.), so $V_{\text{accessible}}(t)$ increases, and therefore $|\mathcal{N}(t)|$ increases.

6.8 Example: Free Expansion of a Gas

Consider a gas initially confined to volume V_0 . At $t = 0$, a partition is removed, and the gas expands to fill volume $V_f > V_0$.

At $t = 0$:

- Actualisation: All molecules in V_0
- Non-actualisations: Molecules distributed in $V_f \setminus V_0$ (not accessible yet)
- $|\mathcal{N}(0)| = 0$ (no alternative distributions are accessible)

At $t = t_f$ (after expansion):

- Actualisation: Molecules uniformly distributed in V_f

- Non-actualisations: All other distributions (e.g., all molecules in left half, all in right half, clustered in corners, etc.)
- $|\mathcal{N}(t_f)| \sim (V_f/V_0)^N$ (number of alternative configurations)

The categorical entropy increase is:

$$\Delta S_{\text{categorical}} = k_B \ln |\mathcal{N}(t_f)| \sim k_B N \ln(V_f/V_0)$$

This matches the kinetic entropy increase from the Sackur-Tetrode equation, showing that categorical and kinetic entropy are closely related for this process.

6.9 Summary

We have established:

1. Non-actualisations are states that could have occurred but did not. They are categorical facts about what did not happen.
2. The information content of an actualisation equals $k_B \ln |\mathcal{N}|$, where $|\mathcal{N}|$ is the number of non-actualisations.
3. Specifying the actualisation requires enumerating all non-actualisations—we must know which states did not occur.
4. Non-actualisations accumulate irreversibly: $d|\mathcal{N}|/dt \geq 0$.
5. Categorical entropy equals the information content of non-actualisations: $S_{\text{categorical}} = k_B \ln |\mathcal{N}|$.

The next section analyzes the logical structure of the set $\mathcal{N}$ and proves that it is self-referential and non-enumerable.

7 Self-Reference and the Impossibility of Complete Enumeration

7.1 The Enumeration Problem

One might object to the preceding arguments: “If we could enumerate all partitions, we could specify the complete state. Then we could reverse it by reversing all partitions simultaneously.”

This section proves that complete enumeration is impossible due to self-reference. The act of enumerating partitions creates new partitions, which themselves require enumeration. This creates an infinite regress that cannot be completed.

7.2 Enumeration Creates Partitions

Theorem 7.1 (Enumeration Creates Partitions). *The act of enumerating a partition $\mathcal{P}$ creates a new partition $\mathcal{P}_{\text{enum}}$ that distinguishes enumerated from non-enumerated states.*

Proof. Suppose we enumerate partition $\mathcal{P} = \{A, \neg A\}$ by listing it explicitly:

$$\text{“Partition } \mathcal{P} \text{ distinguishes } A \text{ from } \neg A\text{”} \quad (101)$$

This enumeration is itself an observation—a physical process that occurs at a specific time and place. The enumeration creates a new distinction:

$$\mathcal{P}_{\text{enum}} = \{\text{enumerated-}\mathcal{P}, \text{not-enumerated-}\mathcal{P}\} \quad (102)$$

Before enumeration:

- State S is characterized by whether $S \in A$ or $S \in \neg A$
- The partition $\mathcal{P}$ exists but has not been enumerated
- $S \in \{\text{not-enumerated-}\mathcal{P}\}$

After enumeration:

- State S is still characterized by whether $S \in A$ or $S \in \neg A$
- The partition $\mathcal{P}$ has been enumerated
- $S \in \{\text{enumerated-}\mathcal{P}\}$

Since the state before enumeration belongs to category $\{\text{not-enumerated-}\mathcal{P}\}$ and the state after enumeration belongs to category $\{\text{enumerated-}\mathcal{P}\}$, these are different states. The enumeration has created a new partition $\mathcal{P}_{\text{enum}}$. $\square$

Concrete example: Consider the partition:

$$\mathcal{P}_{\text{location}} = \{\text{on-table}, \text{not-on-table}\} \quad (103)$$

When you enumerate this partition (write it down, speak it aloud, think about it explicitly), you create:

$$\mathcal{P}_{\text{enum-location}} = \{\text{enumerated-location-partition}, \text{not-enumerated-location-partition}\} \quad (104)$$

Before enumeration, the cup’s state included:

- $\{\text{on-table}\}$ (from $\mathcal{P}_{\text{location}}$)
- $\{\text{not-enumerated-location-partition}\}$ (from $\mathcal{P}_{\text{enum-location}}$)

After enumeration, the cup’s state includes:

- $\{\text{on-table}\}$ (from $\mathcal{P}_{\text{location}}$)
- $\{\text{enumerated-location-partition}\}$ (from $\mathcal{P}_{\text{enum-location}}$)

The state has changed. The enumeration itself is an observation that creates a new partition.

7.3 The Infinite Regress

Theorem 7.2 (Enumeration Infinite Regress). *Attempting to enumerate all partitions creates an infinite regress. Each enumeration creates a new partition that itself requires enumeration.*

Proof. Suppose we attempt to enumerate all partitions. We start with the original partitions:

$$\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_M\} \quad (105)$$

Step 1: Enumerate $\mathcal{P}_1$. This creates:

$$\mathcal{P}_{\text{enum-1}} = \{\text{enumerated-}\mathcal{P}_1, \text{not-enumerated-}\mathcal{P}_1\} \quad (106)$$

Now we have $M + 1$ partitions:

$$\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_M, \mathcal{P}_{\text{enum-1}}\} \quad (107)$$

Step 2: Enumerate $\mathcal{P}_2$. This creates:

$$\mathcal{P}_{\text{enum-2}} = \{\text{enumerated-}\mathcal{P}_2, \text{not-enumerated-}\mathcal{P}_2\} \quad (108)$$

Now we have $M + 2$ partitions:

$$\{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_M, \mathcal{P}_{\text{enum-1}}, \mathcal{P}_{\text{enum-2}}\} \quad (109)$$

Step M : After enumerating all original partitions, we have $2M$ partitions:

$$\{\mathcal{P}_1, \dots, \mathcal{P}_M, \mathcal{P}_{\text{enum-1}}, \dots, \mathcal{P}_{\text{enum-M}}\} \quad (110)$$

But now we must enumerate the enumeration partitions $\{\mathcal{P}_{\text{enum-1}}, \dots, \mathcal{P}_{\text{enum-M}}\}$.

Step $M + 1$: Enumerate $\mathcal{P}_{\text{enum-1}}$. This creates:

$$\mathcal{P}_{\text{enum-enum-1}} = \{\text{enumerated-}\mathcal{P}_{\text{enum-1}}, \text{not-enumerated-}\mathcal{P}_{\text{enum-1}}\} \quad (111)$$

Now we have $2M + 1$ partitions. Continuing this process:

Step $2M$: After enumerating all enumeration partitions, we have $3M$ partitions.

Step $3M$: After enumerating all second-order enumeration partitions, we have $4M$ partitions.

This process never terminates. Each round of enumeration creates M new partitions that themselves require enumeration. The total number of partitions grows without bound:

$$M \rightarrow 2M \rightarrow 3M \rightarrow 4M \rightarrow \dots \rightarrow \infty \quad (112)$$

Therefore, complete enumeration is impossible. $\square$

Analogy: This is similar to Russell’s paradox in set theory. Consider the set of all sets that do not contain themselves:

$$R = \{S : S \notin S\} \quad (113)$$

Does R contain itself? If $R \in R$, then by definition $R \notin R$ (contradiction). If $R \notin R$, then by definition $R \in R$ (contradiction). The paradox arises from self-reference: the definition of R refers to R itself.

Similarly, enumerating partitions creates a self-referential loop: the enumeration of partitions is itself a partition that requires enumeration. This creates an infinite regress that cannot be resolved.

7.4 Gödel’s Incompleteness and Partition Enumeration

The impossibility of complete enumeration is related to Gödel’s incompleteness theorems [15]. Gödel proved that any sufficiently powerful formal system contains true statements that cannot be proved within the system. The proof relies on self-reference: constructing a statement that says “This statement is not provable.”

Our result is analogous:

Theorem 7.3 (Partition Incompleteness). *Any finite enumeration of partitions is incomplete. There always exist partitions (created by the enumeration itself) that are not included in the enumeration.*

Proof. Suppose we have a finite enumeration E of M partitions:

$$E = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_M\} \quad (114)$$

The enumeration E itself creates a partition:

$$\mathcal{P}_E = \{\text{in-enumeration-}E, \text{not-in-enumeration-}E\} \quad (115)$$

This partition $\mathcal{P}_E$ distinguishes partitions that are in E from partitions that are not in E .

Question: Is $\mathcal{P}_E$ in E ?

Case 1: $\mathcal{P}_E \in E$.

Then $\mathcal{P}_E$ was created before the enumeration E was completed. But $\mathcal{P}_E$ is defined by reference to E , so it cannot exist before E is completed. Contradiction.

Case 2: $\mathcal{P}_E \notin E$.

Then E is incomplete—it does not include the partition $\mathcal{P}_E$ created by the enumeration itself.

In both cases, the enumeration E is incomplete. There exists a partition ($\mathcal{P}_E$) that is not included in E . $\square$

Key insight: Just as Gödel showed that formal systems cannot prove all truths about themselves, we show that enumeration systems cannot enumerate all partitions about themselves. Self-reference creates incompleteness.

7.5 The Observer Cannot Enumerate Themselves

A particularly important case of self-reference is the observer attempting to enumerate their own partitions.

Theorem 7.4 (Observer Self-Enumeration Is Impossible). *An observer cannot completely enumerate the partitions that define their own state.*

Proof. Suppose an observer O attempts to enumerate all partitions that define their state. The observer’s state is:

$$O = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \quad (116)$$

The observer begins enumerating: $\mathcal{P}_1, \mathcal{P}_2, \dots$

Each enumeration changes the observer's state by adding a new partition (Theorem 7.1). After enumerating $\mathcal{P}_1$:

$$O' = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \cap \{\text{enumerated-}\mathcal{P}_1\} \quad (117)$$

The observer O' is different from the original observer O . The observer has changed by the act of enumeration.

Continuing this process, after enumerating all M original partitions, the observer has state:

$$O^{(M)} = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \cap \bigcap_{i=1}^M \{\text{enumerated-}\mathcal{P}_i\} \quad (118)$$

But this state includes M new partitions (the enumeration partitions). To completely enumerate the observer's state, these new partitions must also be enumerated. This creates more new partitions, which require enumeration, and so on.

The observer is chasing a moving target: each enumeration changes the observer's state, creating new partitions that require enumeration. The process never completes. $\square$

Analogy: This is like trying to take a photograph of yourself taking a photograph. The photograph includes you holding the camera, but it doesn't include the act of taking that photograph. To include the act of taking the photograph, you need another photograph, which itself requires another act of photographing, and so on. You can never capture the complete state because the act of capturing changes the state.

7.6 Implications for Loschmidt's Reversal

Theorem 7.5 (Loschmidt Requires Complete Enumeration). *To reverse a system from state S_T to state S_0 , one must enumerate all partitions that distinguish S_T from S_0 . By Theorem 7.2, this is impossible.*

Proof. Loschmidt's procedure requires:

1. Identify the initial state S_0 (enumerate all partitions defining S_0)
2. Identify the final state S_T (enumerate all partitions defining S_T)
3. Identify which partitions differ between S_0 and S_T
4. Reverse those partitions to return from S_T to S_0

Step 1 requires enumerating all partitions at t_0 . By Theorem 7.1, this enumeration creates new partitions. By Theorem 7.2, complete enumeration is impossible.

Even if we ignore the enumeration problem, step 3 requires comparing S_0 and S_T . This comparison is itself an observation that creates new partitions:

$$\mathcal{P}_{\text{compare}} = \{\text{same-as-}S_0, \text{different-from-}S_0\} \quad (119)$$

These comparison partitions are not present in S_0 (before the comparison was made). Therefore, even if we successfully reverse all original partitions, the comparison partitions remain, preventing complete return to S_0 . $\square$

Concrete example: Suppose you want to reverse a cup from its current state back to its state 1 hour ago. You must:

1. Remember what the cup looked like 1 hour ago (enumerate partitions at t_0)
2. Observe what the cup looks like now (enumerate partitions at t_T)
3. Compare them (create comparison partitions)
4. Reverse the differences

But the act of remembering, observing, and comparing creates new partitions that were not present 1 hour ago. Even if you successfully reverse the cup's physical state, your memory of the reversal process remains. This memory is a partition that distinguishes the current state from the state 1 hour ago. Therefore, complete reversal is impossible.

7.7 The Measurement Problem in Quantum Mechanics

The self-reference problem is related to the measurement problem in quantum mechanics [?]. In quantum mechanics, a system is described by a wavefunction $|\psi\rangle$ that evolves unitarily (reversibly) according to the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle \quad (120)$$

But when a measurement is made, the wavefunction collapses irreversibly to an eigenstate:

$$|\psi\rangle \rightarrow |\phi_i\rangle \quad (121)$$

This collapse is not described by the Schrödinger equation. It is a separate, irreversible process.

The measurement problem asks: What causes collapse? When does it occur? Is collapse real or apparent?

Our partition framework provides an answer: *Collapse is the creation of a partition*. When a measurement is made, the observer creates a partition:

$$\mathcal{P}_{\text{measurement}} = \{\text{result-}i, \text{not-result-}i\} \quad (122)$$

Before measurement, the system is in a superposition:

$$|\psi\rangle = \sum_i c_i |\phi_i\rangle \quad (123)$$

This superposition means no partition distinguishes the eigenstates $|\phi_i\rangle$. All eigenstates are equally actualized (or equally non-actualized—the distinction doesn't exist yet).

After measurement, the partition $\mathcal{P}_{\text{measurement}}$ is created, and one eigenstate is actualized:

$$\text{Actualized}(\mathcal{P}_{\text{measurement}}) = \{\text{result-}i\} \quad (124)$$

The collapse is not a physical process—it is the creation of a partition that distinguishes one eigenstate from the others. This partition cannot be deleted (by Theorem 5.3), so the collapse is irreversible.

Key insight: The measurement problem arises from trying to describe observation (partition creation) using the same formalism as unobserved evolution (unitary dynamics). These are fundamentally different processes. Unitary evolution preserves information; partition creation adds information. Unitary evolution is reversible; partition creation is irreversible. The measurement problem disappears once we recognize that observation is not a unitary process—it is a partition-creating process.

7.8 The Halting Problem and Partition Enumeration

The impossibility of complete enumeration is also related to the halting problem in computer science [?]. Turing proved that there is no algorithm that can determine, for an arbitrary program and input, whether the program will halt or run forever.

Our result is analogous:

Theorem 7.6 (Partition Enumeration Halting Problem). *There is no algorithm that can enumerate all partitions of a system in finite time.*

Proof. Suppose there exists an algorithm A that enumerates all partitions of a system S in finite time. The algorithm proceeds:

1. Enumerate partition $\mathcal{P}_1$
2. Enumerate partition $\mathcal{P}_2$
3. $\vdots$
4. Enumerate partition $\mathcal{P}_M$
5. Halt (all partitions enumerated)

But by Theorem 7.1, each enumeration creates a new partition. After step M , there are M new partitions (the enumeration partitions) that were not enumerated. Therefore, the algorithm has not enumerated all partitions.

To enumerate the new partitions, the algorithm must continue:

6. Enumerate partition $\mathcal{P}_{\text{enum-1}}$
7. Enumerate partition $\mathcal{P}_{\text{enum-2}}$
8. $\vdots$

But this creates more new partitions, which require enumeration, and so on. The algorithm never halts.

Therefore, no algorithm can enumerate all partitions in finite time. $\square$

Key insight: Just as the halting problem shows that computation has inherent limits (some questions cannot be answered by any algorithm), our result shows that observation has inherent limits (some states cannot be completely enumerated by any observer). These limits are not practical—they are fundamental, arising from self-reference.

7.9 Finite vs. Infinite Enumeration

One might object: “Even if enumeration creates an infinite regress, couldn’t we enumerate all partitions in the limit as time goes to infinity?”

Answer: No, because the enumeration process itself changes the system faster than it can be enumerated.

Theorem 7.7 (Enumeration Divergence). *The rate at which new partitions are created by enumeration exceeds the rate at which partitions can be enumerated. Therefore, the total number of partitions grows without bound.*

Proof. Suppose we enumerate partitions at rate r (partitions per unit time). After time t , we have enumerated:

$$N_{\text{enum}}(t) = rt \quad (125)$$

But each enumeration creates a new partition (Theorem 7.1). Therefore, the total number of partitions is:

$$N_{\text{total}}(t) = N_{\text{original}} + N_{\text{enum}}(t) = N_{\text{original}} + rt \quad (126)$$

The number of unenumerated partitions is:

$$N_{\text{unenumerated}}(t) = N_{\text{total}}(t) - N_{\text{enum}}(t) = N_{\text{original}} \quad (127)$$

Wait, this suggests the number of unenumerated partitions remains constant. But this is incorrect because we have not accounted for second-order enumeration partitions.

More carefully: After enumerating N partitions, we have created N enumeration partitions. These enumeration partitions themselves require enumeration. When we enumerate them, we create N second-order enumeration partitions. And so on.

The total number of partitions after k rounds of enumeration is:

$$N_{\text{total}}(k) = N_{\text{original}} + kN_{\text{original}} = (k+1)N_{\text{original}} \quad (128)$$

This grows linearly with k . But each round of enumeration takes time proportional to the number of partitions enumerated. Therefore, the time required grows quadratically:

$$T(k) \sim k^2 N_{\text{original}} \quad (129)$$

As $k \rightarrow \infty$, the time required diverges faster than the number of partitions enumerated. Therefore, complete enumeration is impossible even in the infinite-time limit. $\square$

Analogy: This is like trying to count all the grains of sand on a beach where each time you count a grain, two new grains appear. You can never finish counting because the sand is appearing faster than you can count it.

7.10 Practical Implications

The impossibility of complete enumeration has practical implications for:

1. State specification in physics:

We can never completely specify the state of a macroscopic system. We can enumerate some partitions (position, momentum, energy, etc.), but there are always additional partitions (enumeration partitions, comparison partitions, observation partitions) that we have not enumerated. This is not a limitation of current technology—it is a fundamental limit arising from self-reference.

2. Reversible computing:

Reversible computing [?] aims to perform computation without entropy increase by using reversible logic gates. But even reversible gates cannot reverse observation. Each observation creates partitions, and these partitions accumulate irreversibly (by Theorem 5.3). Therefore, reversible computing can minimize entropy increase but cannot eliminate it.

3. Information theory:

Shannon's information theory [9] defines information as the reduction of uncertainty. But our results show that observation creates new uncertainty (new partitions that require enumeration) at the same time as it reduces old uncertainty. The total uncertainty never decreases—it only shifts from one form to another.

4. Artificial intelligence:

An AI attempting to model its own state faces the self-enumeration problem (Theorem 7.4). The AI cannot completely model itself because the act of modeling changes the AI's state, creating new partitions that require modeling. This creates an infinite regress that prevents complete self-knowledge.

7.11 Summary: Self-Reference and Enumeration

We have established:

1. **Enumeration creates partitions:** The act of enumerating a partition creates a new partition distinguishing enumerated from non-enumerated states (Theorem 7.1).
2. **Enumeration infinite regress:** Attempting to enumerate all partitions creates an infinite regress—each enumeration creates new partitions that require enumeration (Theorem 7.2).

3. **Partition incompleteness:** Any finite enumeration is incomplete—there always exist partitions not included in the enumeration (Theorem 7.3).
4. **Observer self-enumeration impossible:** An observer cannot completely enumerate the partitions defining their own state (Theorem 7.4).
5. **Loschmidt requires complete enumeration:** Loschmidt's reversal requires enumerating all partitions distinguishing S_T from S_0 , which is impossible (Theorem 7.5).
6. **Connection to Gödel:** Partition incompleteness is analogous to Gödel's incompleteness—self-reference creates fundamental limits on what can be enumerated.
7. **Connection to measurement problem:** Quantum measurement collapse is the creation of a partition, which is irreversible.
8. **Connection to halting problem:** No algorithm can enumerate all partitions in finite time (Theorem 7.6).
9. **Enumeration divergence:** The rate of partition creation exceeds the rate of enumeration, preventing complete enumeration even in the infinite-time limit (Theorem 7.7).

The key insight is: *Observation is self-referential*. The act of observing changes what is being observed, creating new partitions that themselves require observation. This self-reference creates an infinite regress that prevents complete state specification. Therefore, Loschmidt's reversal is impossible—not because we lack sufficient technology or precision, but because the very concept of "complete state specification" is self-contradictory.

8 Loschmidt's Paradox Resolved

8.1 Recapitulation of Loschmidt's Argument

In 1876, Josef Loschmidt posed a challenge to Boltzmann's statistical interpretation of the Second Law [1]. The argument proceeds as follows:

Loschmidt's procedure:

1. Start with a gas in equilibrium at time t_0 (high entropy, uniform distribution)
2. Allow the gas to spontaneously fluctuate to a low-entropy state at time t_T (molecules concentrated in one corner)
3. At time t_T , reverse all molecular velocities: $\mathbf{v}_i \rightarrow -\mathbf{v}_i$
4. The system will evolve backward through its previous trajectory
5. At time $t'_0 = 2t_T - t_0$, the system returns to the initial low-entropy state

6. Entropy has decreased: $S(t'_0) < S(t_T)$

7. This violates the Second Law

Loschmidt argued that since Newtonian mechanics is time-reversible (the equations of motion are invariant under $t \rightarrow -t$, $\mathbf{v} \rightarrow -\mathbf{v}$), this reversal procedure is possible in principle. Therefore, the Second Law cannot be a fundamental law of physics—it must be a statistical tendency that can be violated with sufficient luck or skill.

Boltzmann responded [2] that while the reversal is not impossible, it is astronomically improbable. The probability of spontaneously arriving at the precise reversed state is:

$$P \sim e^{-\Delta S/k_B} \sim e^{-N} \quad (130)$$

where $N \sim 10^{23}$ is the number of molecules. This probability is so small that the reversal would never occur in the age of the universe.

This statistical response has been the standard resolution of Loschmidt's paradox for 150 years. But it is incomplete. We now prove that Loschmidt's reversal is not merely improbable—it is impossible.

8.2 Four Independent Proofs of Impossibility

We have established four independent arguments showing that Loschmidt's reversal is impossible:

Proof 1: Temporal Paradox (Section 2)

Loschmidt's procedure requires both:

- Time progression exists (so the system can evolve from t_0 to t_T)
- Time progression can be undone (so the system can return to t_0)

These requirements are contradictory (Theorem 2.8). If time progression exists, then $t_0 \neq t'_0$ (they are different moments), so the system has not returned to its initial state. If time progression does not exist, there is no evolution to reverse.

Proof 2: Partition Dependence (Section 4)

Observable states are intersections of partition categories (Definition 4.1). The state at t_0 is defined by partitions $\{\mathcal{P}_1, \dots, \mathcal{P}_{M_0}\}$. The state at t'_0 is defined by partitions $\{\mathcal{P}_1, \dots, \mathcal{P}_{M_0}, \mathcal{P}_{M_0+1}, \dots, \mathcal{P}_{M_{0'}}\}$ where $M_{0'} > M_0$ (additional partitions created during evolution and reversal). Since the partition structures are different, the states are different (Theorem 4.2).

Proof 3: Partition Irreversibility (Section 5)

To return from t'_0 to t_0 requires deleting all partitions created during evolution from t_0 to t_T (Theorem 5.4). But partition deletion is impossible because:

- It deletes observed facts (Theorem 5.6)
- It violates causality (Theorem 5.7)
- It requires infinite energy (Theorem 5.8)

- It requires reversing time itself (Theorem 5.11)

Proof 4: Enumeration Self-Reference (Section 7)

To perform the reversal requires enumerating all partitions distinguishing S_T from S_0 (Theorem 7.5). But enumeration creates new partitions (Theorem 7.1), leading to infinite regress (Theorem 7.2). Complete enumeration is impossible (Theorem 7.3).

Each proof is independent and sufficient. Loschmidt's reversal fails for four distinct reasons, any one of which is fatal to the procedure.

8.3 The Fundamental Error in Loschmidt's Argument

Loschmidt's error is assuming that the microstate contains all information about the system. If you know the exact positions and momenta of all molecules at time t_T , you can in principle reverse them and evolve backward.

This assumption is incorrect. The observable state is not just the microstate—it is the microstate plus the partition structure that defines how the microstate is observed.

Theorem 8.1 (Observable State Is Not Microstate). *The observable state S is not identical to the microstate μ . The observable state is the microstate together with the partition structure:*

$$S = (\mu, \{\mathcal{P}_1, \dots, \mathcal{P}_M\}) \quad (131)$$

Proof. By Definition 4.1, an observable state is the intersection of actualized partition categories:

$$S = \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \quad (132)$$

Each partition $\mathcal{P}_i$ divides the space of microstates into two categories. The observable state S is the set of microstates that belong to the actualized category of every partition.

If we specify only the microstate μ (positions and momenta of all particles), we have not specified which partitions are being applied. Different partition structures applied to the same microstate yield different observable states.

Example: Consider a gas with microstate $\mu = \{\mathbf{r}_1, \mathbf{p}_1, \dots, \mathbf{r}_N, \mathbf{p}_N\}$.

Observer 1 applies partitions:

- $\mathcal{P}_{\text{energy}} = \{E \in [E_0, E_0 + \Delta E], E \notin [E_0, E_0 + \Delta E]\}$
- $\mathcal{P}_{\text{volume}} = \{V = V_0, V \neq V_0\}$

and observes: "Gas with energy E_0 in volume V_0 ."

Observer 2 applies partitions:

- $\mathcal{P}_{\text{molecule-1-left}} = \{\text{molecule-1 in left half, molecule-1 in right half}\}$
- $\mathcal{P}_{\text{molecule-2-left}} = \{\text{molecule-2 in left half, molecule-2 in right half}\}$

• $\vdots$

and observes: “Molecule 1 in left half, molecule 2 in right half, ...”

These are different observable states arising from the same microstate μ but different partition structures. Therefore, the observable state is not determined by the microstate alone—it depends on the partition structure. $\square$

Key insight: Loschmidt assumes that reversing the microstate (reversing molecular velocities) is sufficient to reverse the observable state. But this ignores the partition structure. Even if the microstate is reversed, the partition structure is not reversed—it accumulates irreversibly. Therefore, the observable state is not reversed.

8.4 What Loschmidt Actually Reverses

Let us carefully analyze what Loschmidt’s procedure actually accomplishes.

At time t_0 (initial state):

- Microstate: $\mu_0 = \{\mathbf{r}_i(t_0), \mathbf{p}_i(t_0)\}_{i=1}^N$
- Partition structure: $\mathcal{S}_0 = \{\mathcal{P}_1, \dots, \mathcal{P}_{M_0}\}$
- Observable state: $S_0 = (\mu_0, \mathcal{S}_0)$

At time t_T (after evolution):

- Microstate: $\mu_T = \{\mathbf{r}_i(t_T), \mathbf{p}_i(t_T)\}_{i=1}^N$
- Partition structure: $\mathcal{S}_T = \{\mathcal{P}_1, \dots, \mathcal{P}_{M_0}, \mathcal{P}_{M_0+1}, \dots, \mathcal{P}_{M_T}\}$
- Observable state: $S_T = (\mu_T, \mathcal{S}_T)$

Note that $\mathcal{S}_T \supset \mathcal{S}_0$ (the partition structure at t_T includes all partitions from t_0 plus additional partitions created during evolution).

Loschmidt’s reversal at time t_T :

Reverse all molecular velocities:

$$\mathbf{p}_i(t_T) \rightarrow -\mathbf{p}_i(t_T) \quad (133)$$

This creates a new microstate:

$$\mu_T^{\text{rev}} = \{\mathbf{r}_i(t_T), -\mathbf{p}_i(t_T)\}_{i=1}^N \quad (134)$$

But the reversal operation itself is an observation that creates new partitions:

$$\mathcal{P}_{\text{reversal}} = \{\text{velocities-reversed-at-}t_T, \text{velocities-not-reversed-at-}t_T\} \quad (135)$$

Therefore, after reversal:

- Microstate: $\mu_T^{\text{rev}} = \{\mathbf{r}_i(t_T), -\mathbf{p}_i(t_T)\}_{i=1}^N$
- Partition structure: $\mathcal{S}_T^{\text{rev}} = \mathcal{S}_T \cup \{\mathcal{P}_{\text{reversal}}\}$
- Observable state: $S_T^{\text{rev}} = (\mu_T^{\text{rev}}, \mathcal{S}_T^{\text{rev}})$

At time $t'_0 = 2t_T - t_0$ (after reverse evolution):

The microstate evolves backward:

$$\mu_{0'} = \{\mathbf{r}_i(t_0), \mathbf{p}_i(t_0)\}_{i=1}^N = \mu_0 \quad (136)$$

The microstate has returned to its initial value. But the partition structure has not:

$$\mathcal{S}_{0'} = \mathcal{S}_T^{\text{rev}} \cup \{\text{partitions created during reverse evolution}\} \quad (137)$$

Since $\mathcal{S}_{0'} \supset \mathcal{S}_T^{\text{rev}} \supset \mathcal{S}_T \supset \mathcal{S}_0$, we have:

$$\mathcal{S}_{0'} \neq \mathcal{S}_0 \quad (138)$$

Therefore:

$$S_{0'} = (\mu_0, \mathcal{S}_{0'}) \neq (\mu_0, \mathcal{S}_0) = S_0 \quad (139)$$

The observable state has not returned to its initial value, even though the microstate has.

8.5 Entropy Has Not Decreased

Theorem 8.2 (Loschmidt Does Not Decrease Entropy). *Loschmidt’s reversal does not decrease entropy. The entropy at t'_0 is greater than or equal to the entropy at t_T :*

$$S(t'_0) \geq S(t_T) \quad (140)$$

Proof. By Equation 72, entropy is proportional to the number of partitions:

$$S = k_B M \ln 2 \quad (141)$$

At time t_T , the number of partitions is M_T .

At time t'_0 , the number of partitions is:

$$M_{0'} = M_T + 1 + \Delta M \quad (142)$$

where:

- +1 accounts for the reversal partition $\mathcal{P}_{\text{reversal}}$
- ΔM accounts for partitions created during reverse evolution (observations, measurements, environmental interactions)

Since $M_{0'} \geq M_T + 1 > M_T$, we have:

$$S(t'_0) = k_B M_{0'} \ln 2 > k_B M_T \ln 2 = S(t_T) \quad (143)$$

Entropy has increased, not decreased. $\square$

Key point: Even though the microstate at t'_0 is identical to the microstate at t_0 , the entropy at t'_0 is higher than the entropy at t_0 because the partition structure has grown. Entropy is not a property of the microstate—it is a property of the partition structure.

8.6 The Microstate Reversal Is Irrelevant

Loschmidt focuses on reversing the microstate (molecular velocities). But from the perspective of observable states, the microstate reversal is irrelevant. What matters is the partition structure, and the partition structure cannot be reversed.

Theorem 8.3 (Microstate Reversal Does Not Reverse Observable State). *Reversing the microstate does not reverse the observable state because the partition structure is not reversed.*

Proof. The observable state is $S = (\mu, \mathcal{S})$ where μ is the microstate and $\mathcal{S}$ is the partition structure.

Loschmidt's reversal changes:

$$\mu \rightarrow \mu^{\text{rev}} \quad (144)$$

but does not change (in fact, increases):

$$\mathcal{S} \rightarrow \mathcal{S}^{\text{rev}} \supset \mathcal{S} \quad (145)$$

Therefore:

$$\mathcal{S}^{\text{rev}} = (\mu^{\text{rev}}, \mathcal{S}^{\text{rev}}) \neq (\mu, \mathcal{S}) = S \quad (146)$$

The observable state is not reversed, even though the microstate is. $\square$

Analogy: Suppose you write a sentence on a piece of paper: “The cup is on the table.” Then you erase the sentence and write it backward: “elbat eht no si puc ehT.” The letters are reversed, but the paper now has two sentences written on it (one erased, one current). The state of the paper is not reversed—it has changed from “one sentence” to “two sentences (one erased).”

Similarly, Loschmidt reverses the microstate (the “letters”), but the partition structure (the “paper”) accumulates irreversibly. The observable state is not reversed.

8.7 Why Newtonian Mechanics Is Insufficient

Loschmidt's argument relies on the time-reversibility of Newtonian mechanics. The equations of motion:

$$m \frac{d^2 \mathbf{r}_i}{dt^2} = \mathbf{F}_i \quad (147)$$

are invariant under the transformation $t \rightarrow -t$, $\mathbf{v} \rightarrow -\mathbf{v}$. If $\{\mathbf{r}_i(t), \mathbf{v}_i(t)\}$ is a solution, then $\{\mathbf{r}_i(-t), -\mathbf{v}_i(-t)\}$ is also a solution.

This time-reversibility is correct for the microstate dynamics. But it is insufficient to describe observable states, which include the partition structure.

Theorem 8.4 (Newtonian Mechanics Does Not Describe Partition Dynamics). *Newtonian mechanics describes the evolution of the microstate $\mu(t)$ but not the evolution of the partition structure $\mathcal{S}(t)$. Therefore, it cannot determine whether observable states are reversible.*

Proof. Newtonian mechanics provides equations of motion for positions and momenta:

$$\frac{d\mathbf{r}_i}{dt} = \frac{\mathbf{p}_i}{m}, \quad \frac{d\mathbf{p}_i}{dt} = \mathbf{F}_i \quad (148)$$

These equations determine how the microstate $\mu = \{\mathbf{r}_i, \mathbf{p}_i\}$ evolves over time.

But partitions are not dynamical variables in Newtonian mechanics. There are no equations of motion for partitions. Newtonian mechanics does not specify:

- When partitions are created
- Which partitions are created
- How partitions accumulate
- Whether partitions can be deleted

Therefore, Newtonian mechanics is insufficient to determine the evolution of observable states $S = (\mu, \mathcal{S})$. $\square$

Key insight: Newtonian mechanics is a theory of microstate dynamics, not a theory of observation. It tells you how particles move, but not how observations are made or how observable states are defined. To understand thermodynamic irreversibility, we need a theory of observation (partition theory), not just a theory of dynamics (Newtonian mechanics).

8.8 The Role of the Observer

One might object: “But Loschmidt's reversal doesn't require an observer. We just reverse the molecular velocities mechanically, without anyone observing.”

This objection fails because the reversal operation itself creates partitions, independent of whether anyone observes it.

Theorem 8.5 (Reversal Creates Partitions Independent of Observers). *The act of reversing molecular velocities creates partitions in the physical system, independent of whether any observer is present.*

Proof. To reverse molecular velocities, some physical mechanism must:

1. Measure the current velocities $\mathbf{v}_i(t_T)$
2. Apply forces to change velocities to $-\mathbf{v}_i(t_T)$

Step 1 (measurement) creates partitions:

$$\mathcal{P}_{\text{velocity-measured}} = \{\text{velocity-measured-at-}t_T, \text{velocity-not-measured-at-}t_T\} \quad (149)$$

Step 2 (force application) creates partitions:

$$\mathcal{P}_{\text{force-applied}} = \{\text{force-applied-at-}t_T, \text{force-not-applied-at-}t_T\} \quad (150)$$

These partitions are physical facts encoded in the system's state, independent of whether any conscious observer is present. The measurement leaves physical traces (photons scattered, fields perturbed, quantum entanglements created). The force application

leaves physical traces (energy transferred, momentum changed, correlations with force-applying mechanism).

These physical traces are partitions. They distinguish the state after reversal from the state before reversal. Therefore, reversal creates partitions independent of observers. $\square$

Key point: Partitions are not subjective mental constructs—they are objective physical distinctions encoded in the state of the system. Any physical process that creates distinctions (measurement, interaction, correlation) creates partitions. Observers are just one type of partition-creating system. The reversal mechanism itself is a partition-creating system.

8.9 Comparison with Other Resolutions

Our resolution differs from previous approaches:

Boltzmann’s statistical resolution [2]:

Claim: Loschmidt’s reversal is possible but improbable. The probability is $\sim e^{-N}$ where $N \sim 10^{23}$.

Our response: Loschmidt’s reversal is not merely improbable—it is impossible. The probability is not e^{-N} but exactly zero. The reason is not statistical (low probability of arriving at the reversed state) but logical (partition deletion is impossible).

Szilard’s information-theoretic resolution [?]:

Claim: To reverse the system, one must measure the molecular velocities. This measurement creates information, which must be erased. Erasing information generates entropy (by Landauer’s principle [5]), compensating for the entropy decrease.

Our response: This is closer to our resolution, but incomplete. Szilard focuses on information erasure, but does not explain why erasure is necessary or why it generates entropy. Our partition theory provides the explanation: measurement creates partitions, and deleting partitions requires erasing information, which is impossible (or at least requires infinite energy).

Prigogine’s dissipative structures resolution [?]:

Claim: Irreversibility arises from instability and chaos. Small perturbations grow exponentially, making reversal practically impossible due to sensitivity to initial conditions.

Our response: This explains why reversal is practically difficult, but not why it is fundamentally impossible. Our argument shows that even with perfect knowledge and control (no sensitivity to initial conditions), reversal is still impossible due to partition accumulation.

Penrose’s cosmological resolution [3]:

Claim: Time asymmetry arises from special initial conditions—the universe began in a low-entropy state (Big Bang). Entropy increases because we are far from equilibrium, not because of any fundamental irreversibility.

Our response: This explains why entropy is currently increasing, but not why it must always increase. Our

argument shows that entropy increase is a logical necessity arising from partition accumulation, independent of initial conditions.

8.10 The Complete Resolution

We can now state the complete resolution of Loschmidt’s paradox:

Theorem 8.6 (Complete Resolution of Loschmidt’s Paradox). *Loschmidt’s reversal is impossible because:*

1. **Temporal paradox:** *It requires both that time progression exists (for evolution) and that time progression can be undone (for reversal), which are contradictory (Theorem 2.8).*
2. **Partition dependence:** *Observable states are defined by partition structures, not just microstates. The partition structure at t'_0 differs from the partition structure at t_0 , so the observable states differ (Theorem 8.1).*
3. **Partition irreversibility:** *Deleting partitions is impossible because it violates information conservation, causality, and energy conservation (Theorems 5.6, 5.7, 5.8).*
4. **Enumeration self-reference:** *Complete enumeration of partitions is impossible due to self-reference—enumeration creates new partitions that require enumeration (Theorem 7.2).*
5. **Entropy does not decrease:** *Even if the microstate is reversed, entropy increases because the partition structure grows (Theorem 8.2).*

Each reason is independent and sufficient. Loschmidt’s reversal fails for multiple fundamental reasons, not just one.

8.11 Implications for the Second Law

Theorem 8.7 (Second Law Is a Logical Necessity). *The Second Law of Thermodynamics (entropy increases in isolated systems) is a logical necessity, not a statistical tendency or empirical observation.*

Proof. By Theorem 5.9, partitions accumulate irreversibly (they cannot be deleted). By Equation 72, entropy is proportional to the number of partitions:

$$S = k_B M \ln 2 \quad (151)$$

If M increases over time (partitions accumulate), then S increases over time:

$$\frac{dM}{dt} > 0 \implies \frac{dS}{dt} = k_B \ln 2 \frac{dM}{dt} > 0 \quad (152)$$

This is the Second Law. It follows logically from partition irreversibility, not from statistical arguments about the number of microstates. $\square$

Corollary 8.8 (Entropy Increase Is Universal). *All observers, regardless of their physical form or cognitive architecture, observe entropy increase. The Second Law is universal.*

Proof. By Theorem 4.3, all observers agree on which category of a partition is actualized. Therefore, all observers agree on which partitions exist and how many partitions exist. By Equation 72, all observers agree on the entropy. Since partitions accumulate irreversibly for all observers (Theorem 5.3), all observers observe entropy increase. $\square$

8.12 Why Loschmidt's Paradox Persisted for 150 Years

Loschmidt's paradox has persisted for 150 years because the standard resolution (Boltzmann's statistical argument) is incomplete. Boltzmann showed that reversal is improbable, but did not show that it is impossible. This left open the possibility that with sufficient technology or luck, reversal could be achieved.

Our resolution closes this loophole by showing that reversal is not merely improbable but logically impossible. The reasons are:

1. **Confusion of microstate with observable state:** Physicists assumed that the microstate contains all information about the system. This is incorrect—the observable state includes the partition structure, which is not part of the microstate.
2. **Neglect of observation:** Physicists treated observation as passive (merely reading pre-existing information) rather than active (creating partitions that define observable states). This neglect made partition accumulation invisible.
3. **Focus on dynamics rather than structure:** Physicists focused on the dynamics of the microstate (governed by Newton's equations) rather than the structure of observable states (governed by partition accumulation). This made irreversibility appear to be a dynamical property (arising from chaos, instability, etc.) rather than a structural property (arising from partition accumulation).
4. **Lack of formal theory of observation:** Until now, there was no formal theory of observation that could make precise statements about what observation does and why it is irreversible. Partition theory provides this formal framework.

8.13 Summary: Loschmidt's Paradox Resolved

We have established:

1. **Four independent proofs:** Loschmidt's reversal is impossible due to temporal paradox, partition dependence, partition irreversibility, and enumeration self-reference.
2. **Observable state is not microstate:** The observable state includes the partition structure, which is not part of the microstate (Theorem 8.1).

3. **Loschmidt reverses microstate but not observable state:** Reversing molecular velocities reverses the microstate but not the partition structure (Theorem 8.3).
4. **Entropy does not decrease:** Even though the microstate is reversed, entropy increases because the partition structure grows (Theorem 8.2).
5. **Newtonian mechanics is insufficient:** Newtonian mechanics describes microstate dynamics but not partition dynamics (Theorem 8.4).
6. **Reversal creates partitions:** The reversal operation itself creates partitions, independent of observers (Theorem 8.5).
7. **Complete resolution:** Loschmidt's reversal fails for multiple independent reasons (Theorem 8.6).
8. **Second Law is logical necessity:** Entropy increase follows logically from partition irreversibility (Theorem 8.7).
9. **Entropy increase is universal:** All observers observe entropy increase (Corollary 8.8).

The key insight is: *Loschmidt's reversal fails not because it is difficult, but because it is logically contradictory.* The procedure requires deleting partitions (to return to the initial observable state), but partition deletion is impossible (it violates information conservation, causality, and energy conservation). Therefore, the Second Law is not a statistical tendency that could be violated with sufficient luck—it is a logical necessity that cannot be violated under any circumstances.

9 Entropy as Partition Count

9.1 Rederiving the Entropy Formula

We have asserted (Equation 72) that entropy is proportional to the number of partitions:

$$S = k_B M \ln 2 \quad (153)$$

We now derive this formula rigorously from partition theory and show that it is equivalent to the standard Boltzmann entropy formula.

9.2 Boltzmann Entropy

The standard definition of entropy is Boltzmann's formula [2]:

$$S = k_B \ln \Omega \quad (154)$$

where Ω is the number of microstates compatible with the macrostate, and $k_B = 1.380649 \times 10^{-23}$ J/K is Boltzmann's constant.

Interpretation: The macrostate is a coarse-grained description (temperature, pressure, volume). Many different microstates (exact positions and momenta of all particles) correspond to the same macrostate. The entropy measures how many microstates are compatible with the observed macrostate.

Problem: This definition is circular. To compute Ω , we must first define what "compatible with the macrostate" means. This requires specifying which properties of the microstate we are observing (temperature, pressure, volume) and which we are ignoring (exact positions and momenta). This specification is precisely the partition structure.

9.3 Partitions Define Macrostates

Definition 9.1 (Macrostate as Partition Structure). A macrostate is a partition structure $\mathcal{S} = \{\mathcal{P}_1, \dots, \mathcal{P}_M\}$ together with a specification of which category is actualized for each partition.

Example: Ideal gas macrostate.

Partition structure:

$$\mathcal{P}_E = \{E \in [E_0, E_0 + \Delta E], E \notin [E_0, E_0 + \Delta E]\} \quad (155)$$

$$\mathcal{P}_V = \{V = V_0, V \neq V_0\} \quad (156)$$

$$\mathcal{P}_N = \{N = N_0, N \neq N_0\} \quad (157)$$

Actualized categories:

$$\text{Actualized}(\mathcal{P}_E) = \{E \in [E_0, E_0 + \Delta E]\} \quad (158)$$

$$\text{Actualized}(\mathcal{P}_V) = \{V = V_0\} \quad (159)$$

$$\text{Actualized}(\mathcal{P}_N) = \{N = N_0\} \quad (160)$$

Macrostate:

$$\text{Macrostate} = \{E \in [E_0, E_0 + \Delta E]\} \cap \{V = V_0\} \cap \{N = N_0\} \quad (161)$$

This is the set of all microstates with energy in $[E_0, E_0 + \Delta E]$, volume V_0 , and particle number N_0 .

9.4 Microstates Compatible with a Macrostate

Theorem 9.2 (Microstate Count from Partitions). *The number of microstates compatible with a macrostate defined by M partitions is:*

$$\Omega = 2^M \quad (162)$$

assuming each partition divides the microstate space into two equal parts.

Proof. Each partition $\mathcal{P}_i = \{A_i, \neg A_i\}$ divides the space of all microstates into two categories. The macrostate specifies which category is actualized for each partition.

If we have M partitions, there are 2^M possible combinations of actualized categories. Each combination corresponds to a distinct macrostate.

Conversely, if we specify a macrostate (a particular combination of actualized categories), the number of microstates compatible with this macrostate is the size of the intersection:

$$\Omega = \left| \bigcap_{i=1}^M \text{Actualized}(\mathcal{P}_i) \right| \quad (163)$$

If each partition divides the microstate space into two equal parts, and the partitions are independent (the divisions are uncorrelated), then:

$$|\text{Actualized}(\mathcal{P}_i)| = \frac{\Omega_{\text{total}}}{2} \quad (164)$$

where Ω_{total} is the total number of microstates.

After applying M partitions:

$$\Omega = \frac{\Omega_{\text{total}}}{2^M} \quad (165)$$

If we normalize so that $\Omega_{\text{total}} = 2^{M_{\text{max}}}$ (the total microstate space is divided by M_{max} partitions), then:

$$\Omega = 2^{M_{\text{max}} - M} \quad (166)$$

But this counts the number of microstates compatible with a macrostate defined by M partitions. The entropy should depend on M , not on $M_{\text{max}} - M$.

Resolution: We are counting the wrong thing. The entropy should not count the number of microstates compatible with the macrostate—it should count the number of distinguishable macrostates.

With M partitions, we can distinguish 2^M different macrostates (one for each combination of actualized categories). Therefore:

$$\Omega = 2^M \quad (167)$$

□

Key insight: The traditional interpretation of Ω as "number of microstates compatible with the macrostate" is backwards. The correct interpretation is "number of distinguishable macrostates given the partition structure." With M partitions, we can distinguish 2^M macrostates.

9.5 Entropy Formula from Partition Count

Theorem 9.3 (Entropy from Partition Count). *The entropy of a system observed with M partitions is:*

$$S = k_B M \ln 2 \quad (168)$$

Proof. By Theorem 9.2, the number of distinguishable macrostates is:

$$\Omega = 2^M \quad (169)$$

Substituting into Boltzmann's formula (Equation 154):

$$S = k_B \ln \Omega = k_B \ln(2^M) = k_B M \ln 2 \quad (170)$$

□

This is the fundamental entropy formula in partition theory. Entropy is directly proportional to the number of partitions used to observe the system.

9.6 Connection to Information Theory

In information theory [9], the information content of a message is:

$$I = - \sum_i p_i \log_2 p_i \quad (\text{bits}) \quad (171)$$

where p_i is the probability of message i .

For equally probable messages ($p_i = 1/\Omega$ for all i):

$$I = - \sum_{i=1}^{\Omega} \frac{1}{\Omega} \log_2 \frac{1}{\Omega} = -\Omega \cdot \frac{1}{\Omega} \cdot (-\log_2 \Omega) = \log_2 \Omega \quad (172)$$

If $\Omega = 2^M$:

$$I = \log_2(2^M) = M \quad (\text{bits}) \quad (173)$$

Converting to entropy (multiply by $k_B \ln 2$ to convert bits to J/K):

$$S = k_B \ln 2 \cdot I = k_B M \ln 2 \quad (174)$$

This matches Theorem 9.3. Entropy is information content measured in units of J/K instead of bits.

9.7 Why $k_B \ln 2$?

The factor $k_B \ln 2$ converts information (measured in bits) to entropy (measured in J/K).

One bit of information:

$$\Delta S = k_B \ln 2 \approx 9.57 \times 10^{-24} \text{ J/K} \quad (175)$$

This is the entropy increase associated with creating one partition (making one binary distinction).

Why this specific value? The factor k_B sets the energy scale of thermal fluctuations:

$$E_{\text{thermal}} \sim k_B T \quad (176)$$

At room temperature ($T \sim 300 \text{ K}$):

$$E_{\text{thermal}} \sim k_B \cdot 300 \sim 4 \times 10^{-21} \text{ J} \quad (177)$$

The factor $\ln 2$ arises from the binary nature of partitions. Each partition makes a two-way distinction, corresponding to $\ln 2$ nats (or 1 bit) of information.

Together, $k_B \ln 2$ is the entropy cost of making one binary distinction at the thermal energy scale.

9.8 Entropy of an Ideal Gas

We now derive the entropy of an ideal gas using partition theory.

System: N indistinguishable particles in a box of volume V with total energy E .

Microstate: Specified by positions and momenta:

$$\mu = \{\mathbf{r}_1, \mathbf{p}_1, \dots, \mathbf{r}_N, \mathbf{p}_N\} \quad (178)$$

This requires $6N$ continuous parameters.

Macrostate: Specified by coarse-grained observables: E, V, N .

Question: How many partitions are needed to specify the macrostate?

Answer: We must partition both position space and momentum space.

Position partitions:

Divide the volume V into cells of size v_0 (quantum cell volume). The number of cells is:

$$n_V = \frac{V}{v_0} \quad (179)$$

To specify which cell a particle is in requires $\log_2 n_V$ binary partitions. For N particles:

$$M_{\text{position}} = N \log_2 n_V = N \log_2 \frac{V}{v_0} \quad (180)$$

Momentum partitions:

Divide the momentum space into cells of size p_0^3 (quantum momentum cell). The total momentum space volume accessible to the system is determined by the energy constraint:

$$\sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m} = E \quad (181)$$

This defines a hypersphere in $3N$ -dimensional momentum space with radius:

$$p_{\text{total}} = \sqrt{2mE} \quad (182)$$

The volume of this hypersphere is (using the formula for the volume of a $3N$ -dimensional sphere):

$$V_p \sim p_{\text{total}}^{3N} \sim (2mE)^{3N/2} \quad (183)$$

The number of momentum cells is:

$$n_p = \frac{V_p}{p_0^{3N}} \sim \frac{(2mE)^{3N/2}}{p_0^{3N}} \quad (184)$$

To specify which momentum cell the system is in requires:

$$M_{\text{momentum}} = \log_2 n_p = \log_2 \frac{(2mE)^{3N/2}}{p_0^{3N}} = \frac{3N}{2} \log_2 \frac{2mE}{p_0^2} \quad (185)$$

Total partitions:

$$M = M_{\text{position}} + M_{\text{momentum}} = N \log_2 \frac{V}{v_0} + \frac{3N}{2} \log_2 \frac{2mE}{p_0^2} \quad (186)$$

Entropy:

$$S = k_B M \ln 2 = k_B \ln 2 \left(N \log_2 \frac{V}{v_0} + \frac{3N}{2} \log_2 \frac{2mE}{p_0^2} \right) \quad (187)$$

Converting $\log_2$ to $\ln$ (using $\log_2 x = \ln x / \ln 2$):

$$S = k_B \left(N \ln \frac{V}{v_0} + \frac{3N}{2} \ln \frac{2mE}{p_0^2} \right) \quad (188)$$

Quantum cell sizes:

From quantum mechanics, the minimum cell sizes are set by the uncertainty principle:

$$v_0 p_0^3 \sim h^3 \quad (189)$$

where h is Planck's constant. Choosing $v_0 = \lambda^3$ and $p_0 = h/\lambda$ (where λ is the thermal de Broglie wavelength):

$$\lambda = \frac{h}{\sqrt{2\pi m k_B T}} \quad (190)$$

Substituting:

$$S = k_B N \left(\ln \frac{V}{\lambda^3} + \frac{3}{2} \ln \frac{2mE}{h^2/\lambda^2} \right) \quad (191)$$

Simplifying (and using $E = \frac{3}{2} N k_B T$ for an ideal gas):

$$S = k_B N \left(\ln \frac{V}{N \lambda^3} + \frac{5}{2} \right) \quad (192)$$

This is the Sackur-Tetrode equation [?, ?], the standard formula for ideal gas entropy.

9.9 Entropy Increase from Partition Accumulation

Theorem 9.4 (Entropy Increase Rate). *If partitions are created at rate r (partitions per unit time), entropy increases at rate:*

$$\frac{dS}{dt} = k_B r \ln 2 \quad (193)$$

Proof. By Theorem 9.3:

$$S(t) = k_B M(t) \ln 2 \quad (194)$$

If partitions accumulate at rate $r = dM/dt$:

$$\frac{dS}{dt} = k_B \ln 2 \frac{dM}{dt} = k_B r \ln 2 \quad (195)$$

□

Example: Suppose you observe a system once per second, creating one partition per observation. Then $r = 1$ partition/second, and:

$$\frac{dS}{dt} = k_B \ln 2 \approx 9.57 \times 10^{-24} \text{ J/(K} \cdot \text{s)} \quad (196)$$

This is an extremely small entropy increase per observation. But for macroscopic systems with $N \sim 10^{23}$ particles, each observation creates $\sim N$ partitions (one for each particle's state), giving:

$$\frac{dS}{dt} \sim N k_B \ln 2 \sim 10^{23} \times 9.57 \times 10^{-24} \sim 1 \text{ J/(K} \cdot \text{s)} \quad (197)$$

This is a measurable entropy increase.

9.10 Maximum Entropy and Equilibrium

Theorem 9.5 (Maximum Entropy Principle). *A system reaches equilibrium when the number of partitions is maximized subject to constraints (energy, volume, particle number).*

Proof. By Theorem 9.3, entropy is maximized when the number of partitions M is maximized.

The number of partitions is maximized when:

- All possible distinctions have been made (all partitions that can be created have been created)
- No further observations create new partitions (the system is in a stationary state)

This is the definition of equilibrium: a state where no further changes occur, and all macroscopic properties are time-independent.

Therefore, equilibrium corresponds to maximum entropy. □

Example: Gas in a box.

Initially, the gas is concentrated in one corner (low entropy, few partitions distinguish the molecules). Over time, the gas spreads throughout the box (high entropy, many partitions distinguish the molecules). At equilibrium, the gas is uniformly distributed (maximum entropy, maximum number of partitions).

9.11 Entropy and Coarse-Graining

A key insight from partition theory is that entropy depends on the level of coarse-graining—how finely we partition the system.

Theorem 9.6 (Entropy Depends on Coarse-Graining). *The entropy of a system depends on the partition structure used to observe it. Finer partitions (more detailed observations) yield higher entropy.*

Proof. Suppose we observe a system with M_1 partitions, yielding entropy:

$$S_1 = k_B M_1 \ln 2 \quad (198)$$

Now suppose we refine our observations by adding ΔM additional partitions (making finer distinctions). The new entropy is:

$$S_2 = k_B (M_1 + \Delta M) \ln 2 = S_1 + k_B \Delta M \ln 2 > S_1 \quad (199)$$

Therefore, finer observations yield higher entropy. □

Example: Gas in a box.

Coarse observation: Divide the box into 2 halves (left and right). For each particle, create partition:

$$\mathcal{P}_i = \{\text{particle-}i\text{-in-left-half, particle-}i\text{-in-right-half}\} \quad (200)$$

This requires $M_1 = N$ partitions, giving entropy:

$$S_1 = k_B N \ln 2 \quad (201)$$

Fine observation: Divide the box into n cells. For each particle, create $\log_2 n$ partitions to specify which cell it is in. This requires:

$$M_2 = N \log_2 n \quad (202)$$

giving entropy:

$$S_2 = k_B N \log_2 n \ln 2 = k_B N \ln n > S_1 \quad (203)$$

The finer observation yields higher entropy.

Key insight: Entropy is not an intrinsic property of the microstate—it is a property of how the microstate is observed. Different observers using different partition structures observe different entropies for the same microstate.

9.12 Gibbs Paradox Resolution

The Gibbs paradox [?] asks: If we have two identical gases separated by a partition, and we remove the partition, does entropy increase?

Naive answer: Yes. Before removal, each gas has entropy S . After removal, the combined gas has entropy $2S$ (twice as many particles, twice as much volume). But we haven't done anything—we just removed a partition. How can entropy increase?

Standard resolution: The gases are indistinguishable, so we must divide by $N!$ (number of permutations) to avoid overcounting. This gives:

$$S = k_B \ln \frac{\Omega}{N!} \approx k_B (N \ln V - N \ln N + N) \quad (204)$$

When we combine two gases of volume V with N particles each:

$$S_{\text{before}} = 2k_B (N \ln V - N \ln N + N) \quad (205)$$

$$S_{\text{after}} = k_B (2N \ln 2V - 2N \ln 2N + 2N) = 2k_B (N \ln 2V - N \ln 2N + N) \quad (206)$$

Simplifying:

$$S_{\text{after}} - S_{\text{before}} = 2k_B N (\ln 2V - \ln 2N) - 2k_B N (\ln V - \ln N) \stackrel{\text{is:}}{=} 0 \quad (207)$$

No entropy increase. The paradox is resolved.

Partition theory resolution:

Before removal:

- Partition $\mathcal{P}_{\text{left}} = \{\text{in-left-chamber, not-in-left-chamber}\}$ distinguishes left from right
- Each gas is observed separately with M partitions
- Total partitions: $M_{\text{before}} = 1 + 2M$ (one for left/right distinction, M for each gas)

After removal:

- Partition $\mathcal{P}_{\text{left}}$ no longer exists (no barrier distinguishes left from right)
- The combined gas is observed with M' partitions

- Total partitions: $M_{\text{after}} = M'$

If the gases are identical, removing the partition deletes the distinction between left and right. This deletes one partition. But the gases mix, creating new partitions that distinguish different molecular configurations. The net change in partition count depends on whether the mixing creates more partitions than the barrier deletion removes.

For identical gases, the mixing creates no new distinguishable configurations (all molecules are identical, so permuting them doesn't create a new observable state). Therefore:

$$M_{\text{after}} = M_{\text{before}} - 1 + 0 = M_{\text{before}} - 1 \quad (208)$$

Entropy decreases slightly:

$$\Delta S = k_B (M_{\text{after}} - M_{\text{before}}) \ln 2 = -k_B \ln 2 \quad (209)$$

But this is negligible compared to the total entropy ($\sim k_B N \ln 2$ where $N \sim 10^{23}$).

For distinguishable gases (different species), removing the partition allows mixing, which creates new distinguishable configurations. The entropy increases by:

$$\Delta S = k_B N \ln 2 \quad (210)$$

This matches the standard thermodynamic result for mixing entropy.

9.13 Negative Entropy and Information

Schrödinger [?] famously wrote that life "feeds on negative entropy." What does this mean in partition theory?

Answer: Living systems create order (low-entropy structures) by exporting entropy to the environment.

Theorem 9.7 (Entropy Export). *A system can decrease its entropy by transferring partitions to the environment.*

Proof. Consider a system S with M_S partitions and environment E with M_E partitions. The total entropy is:

$$S_{\text{total}} = k_B (M_S + M_E) \ln 2 \quad (211)$$

Suppose the system performs a process that:

- Deletes ΔM partitions from the system
- Creates $\Delta M'$ partitions in the environment

The new total entropy is:

$$S'_{\text{total}} = k_B (M_S - \Delta M + M_E + \Delta M') \ln 2 \quad (212)$$

If $\Delta M' > \Delta M$ (more partitions created in environment than deleted from system):

$$S'_{\text{total}} > S_{\text{total}} \quad (213)$$

Total entropy increases (Second Law satisfied), but system entropy decreases:

$$S'_S = k_B (M_S - \Delta M) \ln 2 < k_B M_S \ln 2 = S_S \quad (214)$$

□

Example: Refrigerator.

The refrigerator decreases entropy inside (cools the interior) by increasing entropy outside (heats the room). The net entropy change is positive:

$$\Delta S_{\text{total}} = \Delta S_{\text{inside}} + \Delta S_{\text{outside}} > 0 \quad (215)$$

even though $\Delta S_{\text{inside}} < 0$.

Similarly, living organisms decrease their internal entropy (create ordered structures like proteins, DNA) by increasing environmental entropy (dissipate heat, excrete waste). The total entropy increases, consistent with the Second Law.

9.14 Summary: Entropy as Partition Count

We have established:

1. **Macrostates are partition structures:** A macrostate is defined by which partitions are applied and which categories are actualized (Definition 9.1).
2. **Microstate count from partitions:** The number of distinguishable macrostates is $\Omega = 2^M$ where M is the number of partitions (Theorem 9.2).
3. **Entropy formula:** Entropy is $S = k_B M \ln 2$ (Theorem 9.3).
4. **Connection to information theory:** Entropy is information content measured in J/K instead of bits.
5. **Ideal gas entropy:** The Sackur-Tetrode equation follows from counting position and momentum partitions.
6. **Entropy increase rate:** If partitions accumulate at rate r , entropy increases at rate $dS/dt = k_B r \ln 2$ (Theorem 9.4).
7. **Maximum entropy principle:** Equilibrium corresponds to maximum partition count (Theorem 9.5).
8. **Entropy depends on coarse-graining:** Finer observations yield higher entropy (Theorem 9.6).
9. **Gibbs paradox resolution:** Removing a partition between identical gases deletes one partition, causing negligible entropy change.
10. **Negative entropy:** Systems can decrease entropy by exporting partitions to the environment (Theorem 9.7).

The key insight is: *Entropy counts partitions, not microstates.* The traditional interpretation of entropy as "number of microstates compatible with the macrostate" is backwards. The correct interpretation is "number of partitions required to specify the macrostate." This makes entropy an observer-dependent quantity—different observers using different partition structures observe different entropies for the same microstate.

10 The Arrow of Time and Cosmological Implications

10.1 The Problem of Time's Arrow

One of the deepest mysteries in physics is the arrow of time: Why does time have a direction? Why do we remember the past but not the future? Why does entropy increase toward the future but not the past?

The fundamental laws of physics—Newton's laws, Maxwell's equations, Schrödinger's equation, Einstein's field equations—are time-symmetric. They work equally well with $t \rightarrow -t$. Yet our experience of time is profoundly asymmetric. We age, memories accumulate, entropy increases—all in one direction.

Previous explanations have appealed to:

- **Statistical mechanics:** Time asymmetry arises from special initial conditions (low-entropy Big Bang) [3]
- **Cosmology:** Time asymmetry is built into the expansion of the universe [?]
- **Quantum mechanics:** Time asymmetry arises from wavefunction collapse [?]
- **Thermodynamics:** Time asymmetry is the Second Law (entropy increases) [?]

Our partition theory provides a new explanation: *Time's arrow is the accumulation of partitions.* Time has a direction because partitions accumulate irreversibly. The past is distinguished from the future by the number of partitions: the future has more partitions than the past.

10.2 Time as Partition Accumulation

Definition 10.1 (Temporal Ordering from Partitions). For two observable states S_1 and S_2 , we say S_1 is earlier than S_2 (written $S_1 \prec S_2$) if the partition structure of S_1 is a subset of the partition structure of S_2 :

$$\mathcal{S}_1 \subseteq \mathcal{S}_2 \quad (216)$$

where $\mathcal{S}_i$ is the partition structure defining state S_i .

Interpretation: The earlier state has fewer partitions than the later state. Time progression is partition accumulation.

Theorem 10.2 (Partition Accumulation Defines Time Direction). *The direction of time is defined by the direction of partition accumulation. If partitions accumulate from S_1 to S_2 (i.e., $\mathcal{S}_1 \subset \mathcal{S}_2$), then S_2 is in the future of S_1 .*

Proof. By Theorem 5.3, partitions cannot be deleted. Therefore, partition structures can only grow or remain constant:

$$\mathcal{S}(t_2) \supseteq \mathcal{S}(t_1) \quad \text{for } t_2 > t_1 \quad (217)$$

If $\mathcal{S}(t_2) \supset \mathcal{S}(t_1)$ (strict superset), then new partitions were created between t_1 and t_2 . These new

partitions distinguish t_2 from t_1 —they encode the fact that time has progressed.

Conversely, if we observe that $\mathcal{S}_2 \supset \mathcal{S}_1$, we can infer that S_2 is later than S_1 (more time has passed, allowing more partitions to accumulate).

Therefore, partition accumulation defines the direction of time. $\square$

Key insight: Time is not a primitive concept—it is defined by partition accumulation. Without partition accumulation, there would be no distinction between past and future, no arrow of time, no temporal ordering of events.

10.3 Why We Remember the Past but Not the Future

Theorem 10.3 (Memory Is Partition Accumulation). *Memory is the accumulation of partitions encoding past observations. We remember the past because those partitions exist. We do not remember the future because those partitions do not yet exist.*

Proof. A memory is a record of a past observation. When you observe an event at time t_1 , you create partitions:

$$\mathcal{P}_{\text{event at } t_1} = \{\text{event-occurred-at-}t_1, \text{event-did-not-occur-at-}t_1\} \quad (218)$$

These partitions are stored in your brain (as neural configurations) and persist over time. At time $t_2 > t_1$, these partitions still exist:

$$\mathcal{S}(t_2) \supset \mathcal{S}(t_1) \supset \{\mathcal{P}_{\text{event at } t_1}\} \quad (219)$$

You can access these partitions (recall the memory) because they are part of your current state.

Conversely, events at time $t_3 > t_2$ (in the future) have not yet occurred. The partitions encoding these events do not yet exist:

$$\mathcal{P}_{\text{event at } t_3} \notin \mathcal{S}(t_2) \quad (220)$$

You cannot remember future events because the partitions encoding them do not exist in your current state.

Therefore, memory asymmetry (remembering past but not future) is a consequence of partition accumulation. $\square$

Analogy: Memory is like a notebook where you write observations. Each observation adds a page. You can read past pages (remember past events) because they exist in the notebook. You cannot read future pages (remember future events) because they haven't been written yet.

The notebook grows over time (partition accumulation). This growth defines the direction of time and explains memory asymmetry.

10.4 Entropy and the Arrow of Time

Theorem 10.4 (Entropy Increase Defines Time Direction). *The direction of time is the direction of entropy increase. If entropy increases from S_1 to S_2 , then S_2 is in the future of S_1 .*

Proof. By Theorem 9.3, entropy is proportional to partition count:

$$S = k_B M \ln 2 \quad (221)$$

By Theorem 10.2, time direction is defined by partition accumulation. If $\mathcal{S}_2 \supset \mathcal{S}_1$, then $M_2 > M_1$, so:

$$S_2 = k_B M_2 \ln 2 > k_B M_1 \ln 2 = S_1 \quad (222)$$

Therefore, entropy increases in the direction of time. Conversely, if we observe entropy increase, we can infer the direction of time. $\square$

Eddington's quote [?]: "The law that entropy always increases holds, I think, the supreme position among the laws of Nature. If someone points out to you that your pet theory of the universe is in disagreement with Maxwell's equations—then so much the worse for Maxwell's equations. If it is found to be contradicted by observation—well, these experimentalists do bungle things sometimes. But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation."

Our partition theory explains why: The Second Law is not just an empirical observation or statistical tendency—it is the definition of time's direction. To violate the Second Law would be to reverse time itself.

10.5 The Big Bang and Initial Conditions

Penrose [3] argued that time's arrow arises from special initial conditions: the universe began in a low-entropy state (Big Bang), and entropy has been increasing ever since. This explains why entropy is currently increasing, but raises the question: Why did the universe begin in a low-entropy state?

Our partition theory provides an answer:

Theorem 10.5 (Initial State Has Minimal Partitions). *The initial state of the universe (Big Bang) has the minimal number of partitions. All partitions are created after the Big Bang through observation and interaction.*

Proof. At the Big Bang ($t = 0$), no observations have yet been made. No partitions have been created. The partition structure is minimal (possibly empty):

$$\mathcal{S}(t = 0) = \emptyset \quad \text{or} \quad \mathcal{S}(t = 0) = \{\text{minimal set of partitions}\} \quad (223)$$

As time progresses, observations occur, interactions happen, and partitions accumulate:

$$\mathcal{S}(t) \supset \mathcal{S}(0) \quad \text{for all } t > 0 \quad (224)$$

By Theorem 9.3, entropy increases:

$$S(t) = k_B M(t) \ln 2 > k_B M(0) \ln 2 = S(0) \quad (225)$$

Therefore, the universe begins in a low-entropy state (minimal partitions) and evolves toward higher entropy (partition accumulation). $\square$

Key insight: The low-entropy initial state is not a special condition that requires explanation—it is the natural consequence of starting with no observations. The universe begins with minimal partitions because no observations have yet been made. Entropy increases because partitions accumulate through observation and interaction.

10.6 Heat Death and Maximum Entropy

The heat death of the universe [?] is the state of maximum entropy, where all energy is uniformly distributed and no further work can be extracted. In partition theory:

Definition 10.6 (Heat Death as Maximum Partitions). Heat death is the state where the partition structure has reached its maximum size. No new partitions can be created because all possible distinctions have been made.

Theorem 10.7 (Heat Death Is Asymptotic). *Heat death is approached asymptotically but never reached in finite time.*

Proof. At any finite time t , the partition structure has finite size:

$$|S(t)| = M(t) < \infty \quad (226)$$

New observations can always create new partitions. For example:

- Observing the system at a new time creates temporal partitions
- Observing at finer resolution creates spatial partitions
- Observing new properties creates property partitions

Therefore, $M(t)$ can always increase:

$$M(t + \Delta t) > M(t) \quad \text{for any } \Delta t > 0 \quad (227)$$

Heat death corresponds to $M \rightarrow \infty$, which is only reached as $t \rightarrow \infty$.

Therefore, heat death is approached asymptotically but never reached in finite time. $\square$

Physical interpretation: Even in a uniform, equilibrium state, there are always quantum fluctuations, thermal noise, and subtle correlations that can be observed. Each observation creates new partitions. Therefore, entropy can always increase, and perfect equilibrium is never achieved.

10.7 Cosmological Expansion and Partition Creation

The expansion of the universe creates new space, which allows new partitions to be created.

Theorem 10.8 (Cosmological Expansion Creates Partitions). *The expansion of the universe increases the number of possible spatial partitions, allowing entropy to increase.*

Proof. At time t , the universe has volume $V(t)$. The number of possible spatial partitions is:

$$M_{\text{spatial}}(t) \sim \log_2 \frac{V(t)}{v_0} \quad (228)$$

where v_0 is the quantum cell volume.

As the universe expands, $V(t)$ increases:

$$V(t_2) > V(t_1) \quad \text{for } t_2 > t_1 \quad (229)$$

Therefore:

$$M_{\text{spatial}}(t_2) > M_{\text{spatial}}(t_1) \quad (230)$$

More spatial partitions can be created, allowing entropy to increase:

$$S(t_2) > S(t_1) \quad (231)$$

$\square$

Key insight: Cosmological expansion is not just the growth of space—it is the creation of new possibilities for partition creation. As the universe expands, more distinctions can be made, more observations can be recorded, more entropy can be generated.

This connects time's arrow to cosmology: the direction of time (partition accumulation) is aligned with the direction of cosmological expansion (volume increase).

10.8 Black Holes and Information Paradox

The black hole information paradox [?] asks: When matter falls into a black hole, does the information it carries disappear? If so, this violates quantum mechanics (which requires information conservation). If not, where does the information go?

Partition theory provides a resolution:

Theorem 10.9 (Black Holes Preserve Partitions). *When matter falls into a black hole, the partitions encoding its state are not destroyed—they are transferred to the black hole's horizon.*

Proof. When matter with partition structure S_{matter} falls into a black hole, the partitions cannot be deleted (by Theorem 5.3). They must go somewhere.

The black hole horizon is a two-dimensional surface encoding information about the interior (holographic principle [?, ?]). The partitions S_{matter} are encoded on the horizon as:

- Perturbations of the horizon geometry

- Quantum entanglement between horizon modes
- Hawking radiation correlations [?]

The black hole's entropy increases by:

$$\Delta S_{\text{BH}} = k_B \Delta M \ln 2 \quad (232)$$

where ΔM is the number of partitions in $\mathcal{S}_{\text{matter}}$.

By the Bekenstein-Hawking formula [?, ?]:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} \quad (233)$$

where A is the horizon area. The increase in entropy corresponds to an increase in horizon area:

$$\Delta A = \frac{4G\hbar}{c^3} \Delta S_{\text{BH}} = \frac{4G\hbar}{c^3} k_B \Delta M \ln 2 \quad (234)$$

Therefore, the partitions are not destroyed—they are transferred to the horizon, increasing its area and entropy. $\square$

Key insight: Black holes do not destroy information—they accumulate partitions on their horizons. The black hole entropy is the number of partitions encoded on the horizon. When the black hole evaporates (Hawking radiation), the partitions are released back into the environment, preserving information.

10.9 Quantum Measurement and Time's Arrow

In quantum mechanics, measurement is irreversible: the wavefunction collapses from a superposition to an eigenstate. This irreversibility is connected to time's arrow.

Theorem 10.10 (Quantum Measurement Creates Partitions). *Quantum measurement creates partitions that distinguish eigenstates. This partition creation is irreversible, defining a direction of time.*

Proof. Before measurement, the system is in a superposition:

$$|\psi\rangle = \sum_i c_i |\phi_i\rangle \quad (235)$$

No partition distinguishes the eigenstates $|\phi_i\rangle$. The partition structure is:

$$\mathcal{S}_{\text{before}} = \{\text{partitions not involving measurement outcome}\} \quad (236)$$

Measurement creates a partition:

$$\mathcal{P}_{\text{measurement}} = \{\text{result-}i, \text{not-result-}i\} \quad (237)$$

After measurement, one eigenstate is actualized:

$$\text{Actualized}(\mathcal{P}_{\text{measurement}}) = \{\text{result-}i\} \quad (238)$$

The partition structure is:

$$\mathcal{S}_{\text{after}} = \mathcal{S}_{\text{before}} \cup \{\mathcal{P}_{\text{measurement}}\} \quad (239)$$

Since $\mathcal{S}_{\text{after}} \supset \mathcal{S}_{\text{before}}$, the measurement defines a direction of time (from before-measurement to after-measurement).

By Theorem 5.3, the partition $\mathcal{P}_{\text{measurement}}$ cannot be deleted. The measurement is irreversible. Therefore, quantum measurement defines an arrow of time. $\square$

Connection to decoherence: In the decoherence interpretation [?], measurement is the entanglement of the system with the environment. This entanglement creates correlations (partitions) that distinguish eigenstates. These correlations spread throughout the environment and cannot be reversed (by Theorem 5.8—reversing requires infinite energy). Therefore, decoherence is irreversible, consistent with our partition theory.

10.10 CPT Theorem and Time Reversal

The CPT theorem [?, ?] states that the combined operation of charge conjugation (C), parity inversion (P), and time reversal (T) is a symmetry of all physical laws. This is often interpreted as: "If you reverse time, flip space, and swap particles with antiparticles, the laws of physics are unchanged."

But the CPT theorem applies to the fundamental laws (Lagrangians, equations of motion), not to observable states. Our partition theory shows that observable states are not CPT-symmetric.

Theorem 10.11 (Observable States Are Not CPT-Symmetric). *Even though the fundamental laws are CPT-symmetric, observable states are not CPT-symmetric because partition structures are not reversed by CPT.*

Proof. The CPT operation acts on the microstate:

$$\text{CPT} : \mu = \{\mathbf{r}_i, \mathbf{p}_i, q_i\} \rightarrow \mu^{\text{CPT}} = \{-\mathbf{r}_i, -\mathbf{p}_i, -q_i\} \quad (240)$$

where q_i is the charge.

But the observable state is $S = (\mu, \mathcal{S})$ where $\mathcal{S}$ is the partition structure. The CPT operation does not act on $\mathcal{S}$:

$$\text{CPT} : S = (\mu, \mathcal{S}) \rightarrow S^{\text{CPT}} = (\mu^{\text{CPT}}, \mathcal{S}) \quad (241)$$

The partition structure is unchanged. Therefore:

$$S^{\text{CPT}} \neq S \quad (242)$$

Observable states are not CPT-symmetric, even though the fundamental laws are. $\square$

Key insight: The CPT theorem tells us that the fundamental laws are time-symmetric, but it does not tell us that observable states are time-symmetric. Time asymmetry arises not from the laws themselves, but from the partition structures that define observable states.

10.11 Retrocausality and Time Travel

Can we send information backward in time? Can we change the past? Partition theory provides definitive answers.

Theorem 10.12 (Retrocausality Is Impossible). *Information cannot be sent backward in time because this would require deleting partitions, which is impossible.*

Proof. Suppose we attempt to send information from time t_2 to time $t_1 < t_2$. At time t_1 , the partition structure is $\mathcal{S}(t_1)$. At time t_2 , the partition structure is $\mathcal{S}(t_2) \supset \mathcal{S}(t_1)$.

To send information backward means to create a partition at t_1 that encodes information from t_2 :

$$\mathcal{P}_{\text{info from } t_2} \in \mathcal{S}(t_1) \quad (243)$$

But this partition did not exist at t_1 (it encodes information from t_2 , which had not yet occurred). To create it at t_1 requires changing the past—making $\mathcal{S}(t_1)$ different from what it actually was.

By Theorem 5.7, changing the past violates causality. Therefore, retrocausality is impossible. $\square$

Corollary 10.13 (Time Travel to the Past Is Impossible). *Time travel to the past is impossible because it would require deleting all partitions created between the present and the past.*

Proof. To travel from time t_2 to time $t_1 < t_2$ requires returning to the state at t_1 . This requires:

$$\mathcal{S}(t_2) \rightarrow \mathcal{S}(t_1) \quad (244)$$

Since $\mathcal{S}(t_2) \supset \mathcal{S}(t_1)$, this requires deleting partitions:

$$\Delta\mathcal{S} = \mathcal{S}(t_2) \setminus \mathcal{S}(t_1) \quad (245)$$

By Theorem 5.3, partition deletion is impossible. Therefore, time travel to the past is impossible. $\square$

Note: Time travel to the future is possible (in the sense of experiencing less proper time than the external universe) via time dilation in special relativity. But this is not true time travel—it is just differential aging. The partition structure still accumulates monotonically along the traveler's worldline.

10.12 The Beginning and End of Time

Theorem 10.14 (Time Begins with First Partition). *Time begins when the first partition is created. Before the first partition, there is no temporal ordering, no distinction between past and future.*

Proof. By Definition 10.1, temporal ordering is defined by partition accumulation. If no partitions exist, no temporal ordering exists.

At the Big Bang, $\mathcal{S}(t = 0) = \emptyset$ (or minimal). The first partition is created at $t = t_1 > 0$:

$$\mathcal{S}(t_1) = \{\mathcal{P}_1\} \quad (246)$$

This partition distinguishes t_1 from $t_0 = 0$. It creates the first temporal distinction, defining the beginning of time.

Before t_1 , no temporal distinctions exist. There is no meaningful sense in which $t = 0$ is "before" $t = t_1$ —these are just labels. Time only begins when partitions begin. $\square$

Theorem 10.15 (Time Ends with Last Partition). *If partition creation ceases (no new partitions are created), time effectively ends. There is no temporal progression without partition accumulation.*

Proof. By Theorem 10.2, time progression is partition accumulation. If $\mathcal{S}(t_2) = \mathcal{S}(t_1)$ for all $t_2 > t_1$, then no temporal distinction exists between t_1 and t_2 . Time has stopped.

This could occur in the heat death scenario (Definition 10.6), where all possible partitions have been created and no new partitions can be formed. At this point, the universe reaches a static state with no temporal progression. $\square$

Philosophical implication: Time is not a container in which events occur—it is the accumulation of events (partitions). Without events, there is no time. The beginning of time is the first event. The end of time is the last event.

10.13 Multiple Arrows of Time

Various physical processes define "arrows of time":

- **Thermodynamic arrow:** Entropy increases
- **Cosmological arrow:** Universe expands
- **Psychological arrow:** We remember past, not future
- **Causal arrow:** Causes precede effects
- **Electromagnetic arrow:** Radiation propagates outward from sources
- **Quantum arrow:** Wavefunction collapses, not un-collapses

Are these independent arrows, or are they all manifestations of a single underlying arrow?

Theorem 10.16 (All Arrows Are Partition Accumulation). *All arrows of time are manifestations of partition accumulation. They all point in the same direction because they all arise from the irreversibility of partition creation.*

Proof. **Thermodynamic arrow:** By Theorem 10.4, entropy increase is partition accumulation.

Cosmological arrow: By Theorem 10.8, expansion creates new spatial partitions.

Psychological arrow: By Theorem 10.3, memory is partition accumulation.

Causal arrow: Causation is the creation of correlations (partitions) between cause and effect. Causes precede effects because the correlation partitions are created after the cause.

Electromagnetic arrow: Radiation carries information (partitions) outward from sources. The outward propagation is the spreading of partitions through space.

Quantum arrow: By Theorem 10.10, wavefunction collapse is partition creation.

All arrows involve partition creation or accumulation. Since partition creation is irreversible (Theorem 5.3), all arrows point in the same direction. $\square$

Key insight: There is only one arrow of time—partition accumulation. All other arrows are manifestations of this fundamental arrow in different physical contexts.

10.14 Summary: Arrow of Time and Cosmology

We have established:

1. **Time direction from partitions:** Time’s direction is defined by partition accumulation (Theorem 10.2).
2. **Memory asymmetry:** We remember the past because past partitions exist; we don’t remember the future because future partitions don’t yet exist (Theorem 10.3).
3. **Entropy defines time:** The direction of entropy increase is the direction of time (Theorem 10.4).
4. **Big Bang initial conditions:** The universe begins with minimal partitions, naturally explaining low initial entropy (Theorem 10.5).
5. **Heat death:** Maximum entropy (maximum partitions) is approached asymptotically (Theorem 10.7).
6. **Cosmological expansion:** Expansion creates new spatial partitions, allowing entropy to increase (Theorem 10.8).
7. **Black hole information:** Black holes preserve partitions on their horizons (Theorem 10.9).
8. **Quantum measurement:** Measurement creates partitions, defining time’s arrow (Theorem 10.10).
9. **CPT asymmetry:** Observable states are not CPT-symmetric because partition structures are not reversed (Theorem 10.11).
10. **No retrocausality:** Information cannot be sent backward in time (Theorem 10.12).
11. **No time travel:** Travel to the past is impossible (Corollary 10.13).
12. **Beginning and end of time:** Time begins with the first partition and ends when partition creation ceases (Theorems 10.14, 10.15).
13. **Unified arrow:** All arrows of time are manifestations of partition accumulation (Theorem 10.16).

The key insight is: *Time is partition accumulation*. The arrow of time is not a mysterious asymmetry imposed on time-symmetric laws—it is the logical consequence of partition irreversibility. Time flows forward because partitions accumulate irreversibly. The past is distinguished from the future by the number of partitions. This explains all temporal asymmetries in physics, from entropy increase to memory to causation.

11 Conclusion

We have resolved Loschmidt’s paradox by demonstrating that entropy reversal is logically impossible. This conclusion follows from the self-referential structure of non-actualisations, which renders them non-enumerable in standard set theory.

11.1 Summary of Main Results

The resolution proceeds through six key steps:

1. Decomposition of entropy (Section 3): Total entropy comprises two independent components:

$$S_{\text{total}} = S_{\text{kinetic}} + S_{\text{categorical}}$$

Kinetic entropy measures energy dispersal across degrees of freedom. Categorical entropy measures the information content of distinctions—the number of ways state space has been partitioned into distinguishable categories.

2. Partition lag mechanism (Section ??): Every partition operation has an irreducible temporal gap $\tau_{\text{lag}} > 0$ between initiation and completion. During this interval, the system evolves, creating an undetermined residue $\mathcal{R}$ —information about states visited during $[t_{\text{start}}, t_{\text{end}})$ that is permanently inaccessible. This generates categorical entropy:

$$\Delta S_{\text{categorical}} = k_B \ln |\mathcal{R}|$$

3. Irreversibility of categorical entropy: Partition operations cannot be inverted. They form a semigroup (closed under composition) but not a group (no inverses). Each partition generates entropy that cannot be removed by subsequent partitions.

4. Non-actualisations accumulate monotonically (Section 6): Non-actualisations are states that could have occurred but did not. They are categorical facts about what did not happen. The set of non-actualisations $\mathcal{N}(t)$ grows monotonically:

$$\frac{d|\mathcal{N}|}{dt} \geq 0$$

This growth is irreversible because past non-actualisations remain non-actualisations—the fact that a state did not occur at time t is permanent.

5. Self-reference in non-actualisations (Section ??): Every non-actualisation “not X ” is simultaneously an actualisation $\neg X$ of the complementary state. This creates self-reference analogous to Russell’s paradox. The set $\mathcal{N}$ is not well-defined in Zermelo-Fraenkel set theory and is not enumerable. There does not exist a surjective function $f : \mathbb{N} \rightarrow \mathcal{N}$.

6. Impossibility of Loschmidt’s reversal (Section ??): Specifying the microstate $\mathbf{x}(t)$ required for velocity inversion is equivalent to enumerating $\mathcal{N}(t)$. Since $\mathcal{N}(t)$ is non-enumerable, $\mathbf{x}(t)$ cannot be specified completely. Therefore, Loschmidt’s reversal cannot be performed. This is a logical impossibility, not a practical limitation.

11.2 The Nature of the Resolution

Our resolution differs fundamentally from previous approaches in three respects:

Logical rather than statistical: Standard resolutions treat entropy increase as overwhelmingly probable but not certain. Our resolution shows it is logically necessary. The non-enumerability of non-actualisations is a theorem about self-referential sets, independent of probability or statistics.

Absolute rather than observer-dependent: Coarse-graining approaches make entropy depend on the observer’s resolution or knowledge. Our resolution shows that categorical entropy is objective—it depends on which states actually occurred, not on what observers know about them.

Intrinsic rather than environmental: Decoherence approaches attribute irreversibility to environmental entanglement. Our resolution shows that irreversibility is intrinsic to partition operations themselves, independent of whether an environment is present.

The key insight is that negation has structure. To say “ X did not occur” is not merely to note an absence—it is to assert a fact about the universe. This fact is itself part of the universe’s state. Therefore, every non-actualisation is simultaneously an actualisation of a complementary state. This self-reference makes the set of non-actualisations non-enumerable, which in turn makes entropy reversal logically impossible.

11.3 Why Previous Approaches Were Incomplete

Previous resolutions of Loschmidt’s paradox addressed kinetic entropy but overlooked categorical entropy.

Boltzmann’s statistical approach: Boltzmann showed that low-entropy states are rare in phase space. A system in a low-entropy state will almost certainly evolve toward higher entropy because there are vastly more high-entropy microstates. However, this leaves open the logical possibility of entropy decrease—it is merely improbable, not impossible. Moreover, it does not explain why low-entropy states are rare in the first place (the problem of initial conditions).

Gibbs’s coarse-graining approach: Gibbs distinguished fine-grained entropy (which is conserved by Hamiltonian dynamics) from coarse-grained entropy (which increases). Coarse-grained entropy depends on how we partition phase space into macroscopic cells. This makes entropy observer-dependent: different coarse-grainings yield different entropies. Our framework shows that categorical entropy is not arbitrary coarse-graining—it is determined by the partition operations that actually occur in the system’s dynamics.

Quantum decoherence: Zurek and others showed that environmental entanglement destroys quantum coherence, making reversed trajectories physically inaccessible even if they are formally allowed by unitary evolution. However, this applies only to open quantum

systems with environments. Our framework shows that irreversibility is intrinsic to partition operations, which occur even in isolated systems (through internal dynamical distinctions).

Cosmological boundary conditions: Penrose and others attribute the arrow of time to the universe’s low-entropy initial state at the Big Bang. This explains why entropy increases globally but does not explain why local entropy reversals (like Loschmidt’s reversal) are impossible. Our framework shows that local reversals fail due to the non-enumerability of non-actualisations, independent of cosmological boundary conditions.

All these approaches are correct within their domains but incomplete. They address kinetic entropy (energy dispersal) but not categorical entropy (partition structure). Our framework completes the picture by showing that categorical entropy is irreversible due to the self-referential structure of non-actualisations.

11.4 The Second Law as a Logical Principle

Our results elevate the Second Law of Thermodynamics from an empirical regularity to a logical principle.

Historically, the Second Law was discovered empirically: heat flows from hot to cold, gases expand to fill containers, ordered structures decay. Clausius formulated it as $\Delta S \geq 0$ for isolated systems. Boltzmann provided a statistical explanation: entropy increase is probable because high-entropy states are numerous.

Our framework shows that the Second Law is not merely probable—it is logically necessary. Entropy must increase because:

1. Every physical process involves partition operations (measurements, collisions, energy exchanges).
2. Every partition operation generates categorical entropy $\Delta S_{\text{categorical}} = k_B \ln |\mathcal{R}|$ due to partition lag.
3. Categorical entropy cannot be reversed because non-actualisations are non-enumerable.
4. Therefore, total entropy $S_{\text{total}} = S_{\text{kinetic}} + S_{\text{categorical}}$ increases monotonically.

This chain of reasoning is logical, not statistical. It does not depend on probabilities, large numbers, or typical behavior. It depends only on the structure of negation and the self-reference inherent in non-actualisations.

This does not mean the Second Law can be derived from pure logic alone—it requires physical premises (that systems evolve, that partition operations occur, that states are distinguishable). But given these premises, the conclusion is logically necessary.

11.5 Relation to Gödel and Russell

The non-enumerability of non-actualisations is structurally analogous to Gödel’s incompleteness theorem and Russell’s paradox.

Russell’s paradox: The set $R = \{S : S \notin S\}$ (sets that do not contain themselves) is not well-defined because asking whether $R \in R$ leads to contradiction. The resolution is that R cannot be formed in Zermelo-Fraenkel set theory due to the axiom of foundation.

Gödel’s theorem: In any consistent formal system F capable of expressing arithmetic, there exist true statements that cannot be proven within F . The statement “this statement is unprovable in F ” is true but unprovable. The resolution is that no formal system can prove all truths about arithmetic.

Non-actualisation theorem: The set $\mathcal{N} = \{X : X \text{ did not occur}\}$ is not well-defined because each element “not X ” is simultaneously an actualisation $\neg X$. Asking whether “being in $\mathcal{N}$ ” occurred leads to contradiction. The resolution is that $\mathcal{N}$ cannot be enumerated.

All three results establish fundamental limits on formal systems:

- Russell: Not all sets can be formed.
- Gödel: Not all truths can be proven.
- Non-actualisation theorem: Not all non-actualisations can be enumerated.

All three limits arise from self-reference:

- Russell: A set referring to its own membership.
- Gödel: A statement referring to its own provability.
- Non-actualisation theorem: A state referring to its own non-occurrence.

This suggests a deep connection between logic, mathematics, and physics. The structure of negation imposes constraints not only on formal systems (Russell, Gödel) but also on physical systems (non-actualisation theorem). The impossibility of entropy reversal is a physical manifestation of the same self-reference that makes Russell’s set ill-defined and Gödel’s statement unprovable.

11.6 Philosophical Implications

Our results have implications for the philosophy of time, causation, and physical law.

The arrow of time: We have shown that the arrow of time is absolute and intrinsic. It is not imposed by observers (contra coarse-graining approaches), not contingent on environmental decoherence (contra quantum approaches), and not dependent on cosmological boundary conditions (contra Penrose). The arrow emerges from the structure of partition operations: time flows in the direction of increasing categorical entropy, which is determined by the accumulation of non-actualisations. This accumulation is irreversible due to the non-enumerability of $\mathcal{N}$.

Determinism and irreversibility: Classical mechanics is deterministic: given initial conditions, the future is uniquely determined. Yet entropy increase is

irreversible. How can a deterministic theory produce irreversible behavior? Our answer: determinism applies to actualisations (what happens), but irreversibility arises from non-actualisations (what does not happen). The set of non-actualisations grows monotonically even though the actual trajectory is deterministic. Irreversibility is not a property of the trajectory itself but of the context in which it is embedded—the space of alternative trajectories that did not occur.

The status of physical laws: Are the laws of thermodynamics fundamental or emergent? Our framework suggests they are neither. The Second Law is not fundamental in the sense of being a basic postulate (like Newton’s laws or the Schrödinger equation). But it is not emergent in the sense of arising from statistical behavior of large systems. Instead, it is a logical consequence of the structure of partition operations. It is a meta-law—a constraint on what physical laws can permit, arising from the logical structure of negation.

11.7 Open Questions

Several questions remain for future investigation:

1. Quantum non-actualisations: In quantum mechanics, non-actualisations correspond to measurement outcomes that did not occur (e.g., $|\downarrow\rangle$ when $|\uparrow\rangle$ was measured). Do these form a self-referential set analogous to the classical case? The structure may differ because quantum superposition allows states to be “partially actualized” before measurement. A full treatment would require analyzing the self-reference structure of the quantum measurement problem.

2. Relation to computational complexity: The non-enumerability of $\mathcal{N}$ suggests connections to computational complexity theory. Is there a complexity class corresponding to the difficulty of enumerating non-actualisations? Does this relate to the P vs NP problem or to oracle hierarchies in computability theory?

3. Alternative set theories: We have shown that $\mathcal{N}$ is not well-defined in Zermelo-Fraenkel set theory. Are there alternative set-theoretic foundations (e.g., non-well-founded set theory, category theory) in which $\mathcal{N}$ can be consistently defined? If so, would this allow entropy reversal in those frameworks?

4. Experimental tests: Are there experimental consequences of categorical entropy that differ from kinetic entropy? For example, does categorical entropy contribute to gravitational mass (via $E = mc^2$)? Can we measure partition lag directly in quantum systems with ultrafast spectroscopy?

5. Generalization to other irreversible processes: Does the non-enumerability of non-actualisations explain other irreversible phenomena (e.g., wavefunction collapse, CP violation in particle physics, the cosmological arrow of time)? Is there a unified framework in which all irreversibility arises from self-reference in non-actualisations?

11.8 Final Remarks

Loschmidt's paradox has stood for 150 years as a challenge to the foundations of statistical mechanics. The paradox is sharp: if the microscopic laws are time-reversible, why is the macroscopic world irreversible?

We have resolved this paradox by showing that entropy reversal requires enumerating non-actualisations, which is logically impossible due to self-reference. The impossibility is absolute—it holds regardless of technological capabilities, measurement precision, or computational power. It is a consequence of the structure of negation itself.

This resolution reveals that irreversibility is not a statistical accident, not an observer convention, not a quantum phenomenon, and not a cosmological contingency. It is a logical necessity arising from the self-referential structure of non-actualisations.

The Second Law of Thermodynamics is thus elevated from an empirical regularity to a logical principle. Entropy must increase because the set of non-actualisations is non-enumerable, and this non-enumerability is a theorem about self-referential sets.

In this sense, the arrow of time is not imposed on the universe from outside—it is intrinsic to the logical structure of physical states. Time flows forward because negation has structure, and that structure is self-referential.

Loschmidt asked: If the laws are reversible, why is the world irreversible? Our answer: Because to reverse the world, we must enumerate what did not happen. And what did not happen cannot be enumerated, because it refers to itself.

The paradox is resolved. Entropy increase is not merely probable—it is logically necessary.

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